

# Lecture 5: Image Transforms and Image Enhancement in Frequency Domain



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thanks to G&W website, Lexing Xie, and Alan Peters for slide materials

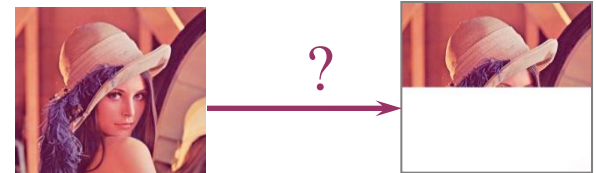
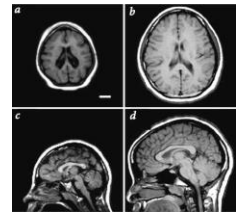


# Outline

- Why transform?
- 2D Fourier transform
  - DFT and 2D-DFT
  - properties
  - implementation
  - applications
- Discrete cosine transform (DCT)
  - definition & visualization
  - Implementation
- Multi-resolution and wavelets

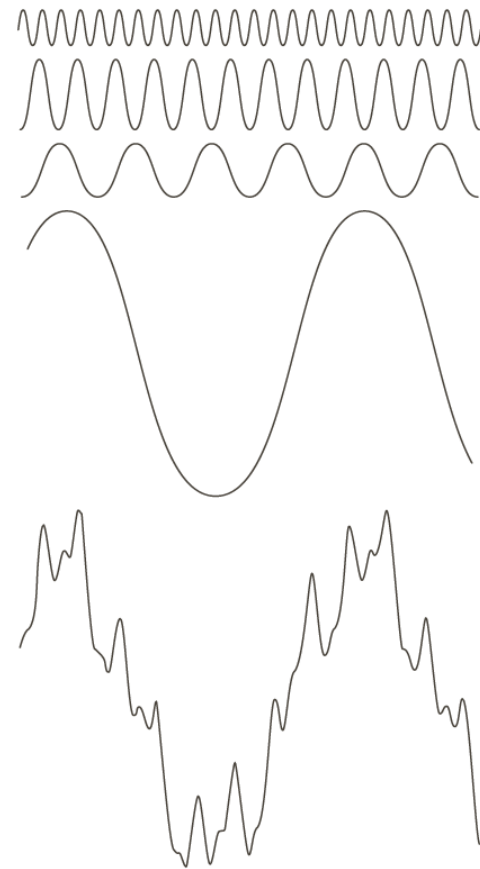
# Why do we need image transform?

- Better image processing
  - Take into account long-range correlations in space
  - Conceptual insights in spatial-frequency information.  
what it means to be “smooth, moderate change, fast change, ...”
- Fast computation: convolution vs. multiplication
- Alternative representation and sensing
  - Obtain transformed data as measurement in radiology images (medical and astrophysics), inverse transform to recover image
- Efficient storage and transmission
  - Energy compaction
  - Pick a few “representatives” (basis)
  - Just store/send the “contribution” from each basis



# Fourier Transform

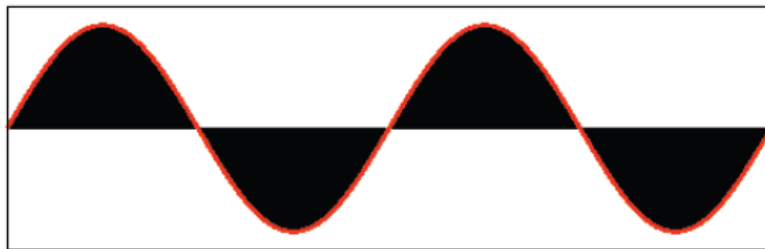
- Fourier transform: a continuous signal can be represented as a (countable) weighted sum of sinusoids.
- The sinusoids are called “basis functions”.



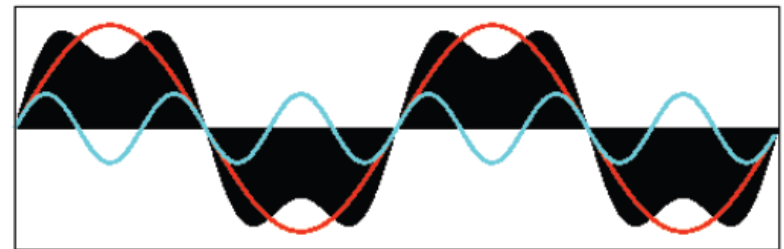
**FIGURE 4.1** The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

# Square Wave

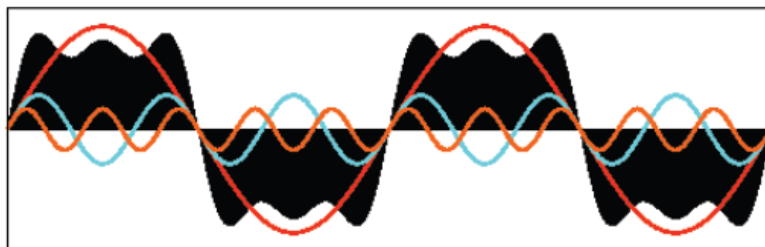
Frequencies:  $f$



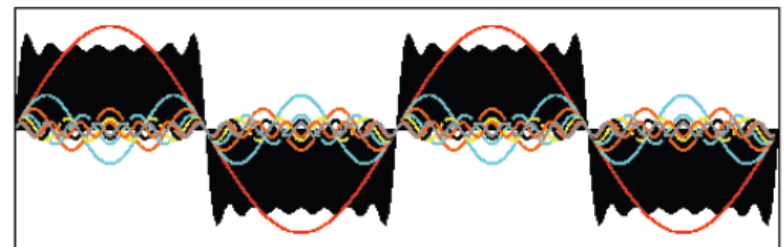
Frequencies:  $f + 3f$



Frequencies:  $f + 3f + 5f$

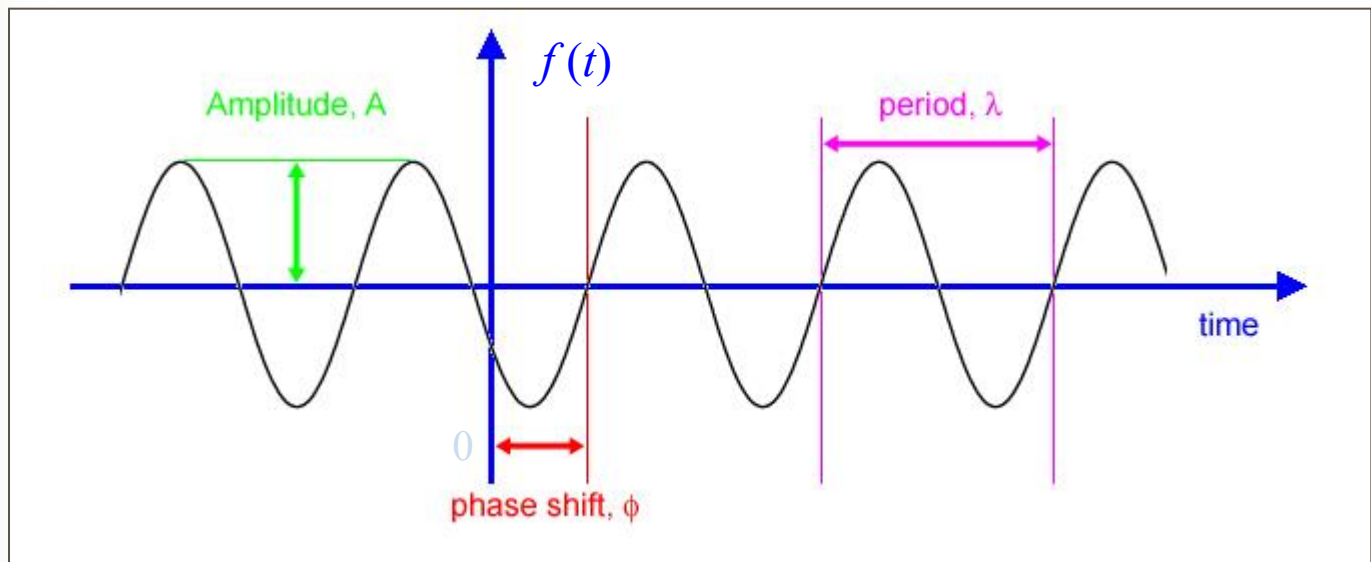


Frequencies:  $f + 3f + 5f + \dots + 15f$





# Anatomy of a Sinusoid



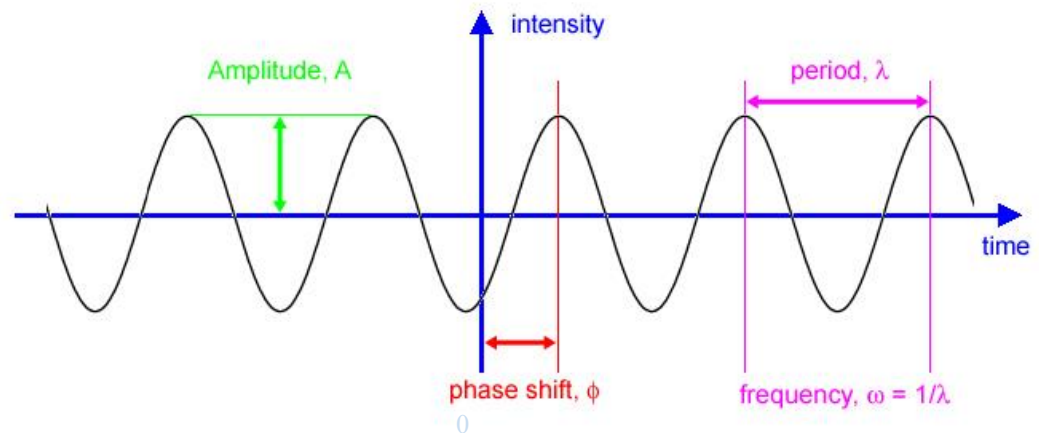
$$f(t) = A \sin\left(\frac{2\pi}{\lambda} t - \phi\right)$$

$1/\lambda$  is the frequency of the sinusoid (Hz).  
 $2\pi/\lambda$  is the angular frequency (radians/s).

## Fourier Coefficients Are Complex Numbers

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{+i \frac{2\pi n}{\lambda} t} \quad \text{where} \quad C_n = |C_n| e^{+i \phi_n}.$$

$C_n$  represents the amplitude,  $A=|C_n|$ , and relative phase,  $\phi$ , of that part of the original signal,  $f(t)$ , that is a sinusoid of frequency  $\omega_n = n / \lambda$ .



$n$  is the integer sequence 1,2,3,..., as frequency index

# The Fourier Transform

is the decomposition of a *nonperiodic* signal into a continuous sum\* of sinusoids.

$$\begin{aligned} F(\omega) &= |F(\omega)| e^{i\Phi(\omega)} = \int_{-\infty}^{\infty} f(t) e^{i2\pi\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) [\cos(2\pi\omega t) + i \sin(2\pi\omega t)] dt \end{aligned}$$

Inverse transform:

$$\begin{aligned} f(t) &= \int_{-\infty}^{\infty} F(\omega) e^{-i2\pi\omega t} d\omega = \int_{-\infty}^{\infty} |F(\omega)| e^{-i(2\pi\omega t + \Phi(\omega))} d\omega \\ &= \int_{-\infty}^{\infty} F(\omega) [\cos(2\pi\omega t) - i \sin(2\pi\omega t)] d\omega \\ &= \int_{-\infty}^{\infty} |F(\omega)| [\cos(2\pi\omega t + \Phi(\omega)) - i \sin(2\pi\omega t + \Phi(\omega))] d\omega \end{aligned}$$

\* *i.e.*, an integral.



# 1-D continuous FT

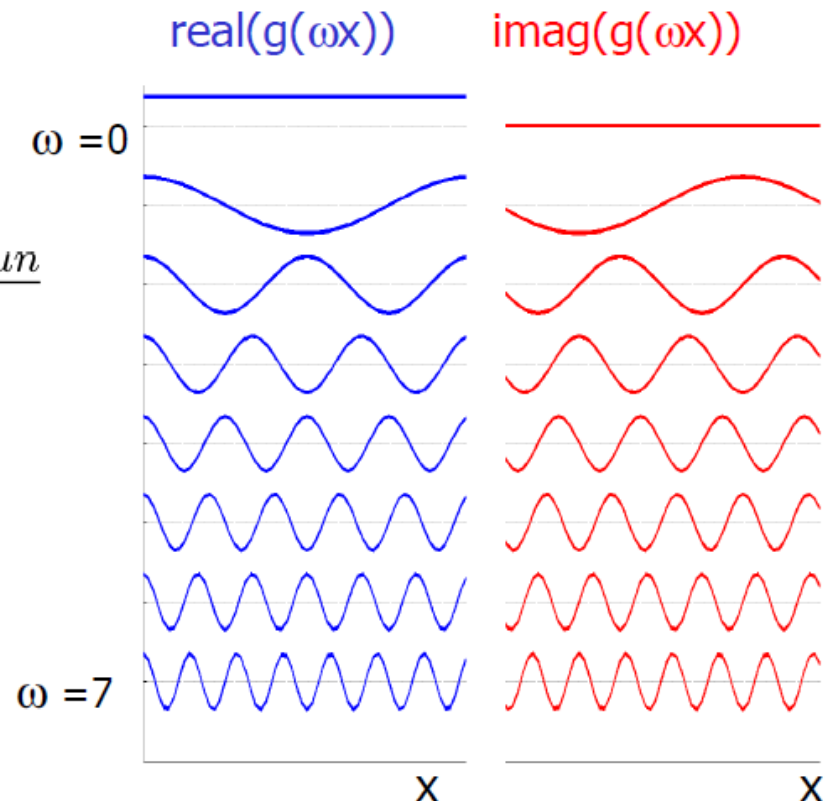
- 1D – FT

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-j2\pi\omega x} dx$$

$$g(\omega x) = e^{-j2\pi\omega x}$$

- 1D – DFT of length N

$$F(u) = \frac{1}{N} \sum_{n=0}^{N-1} f(n) e^{-\frac{j2\pi un}{N}}$$



# 1-D DFT in as basis expansion

$$F(u) = \frac{1}{N} \sum_{n=0}^{N-1} f(n) e^{-j2\pi \frac{un}{N}}$$

Forward transform

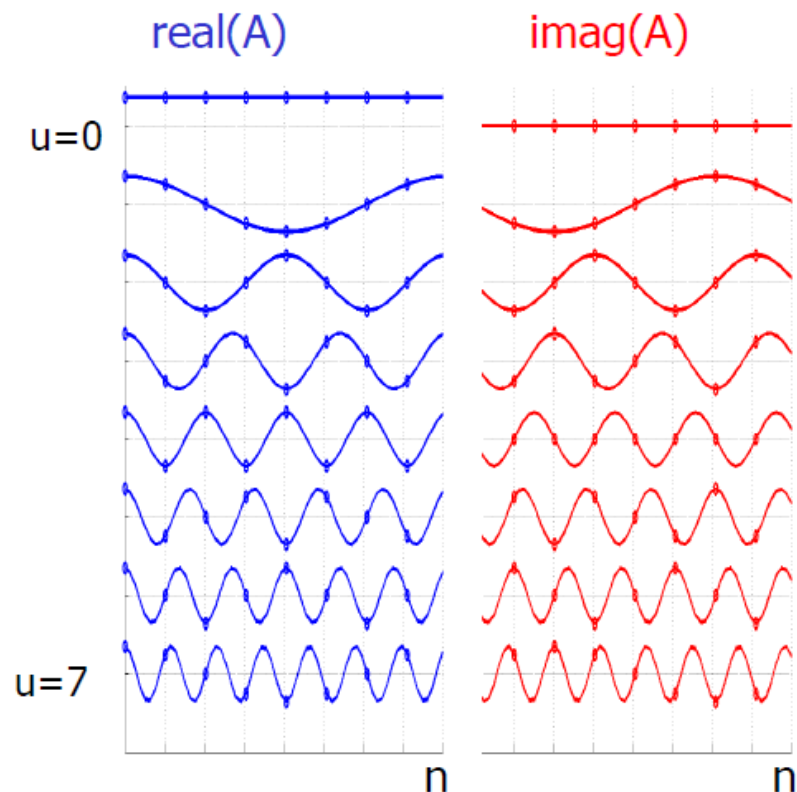
$$y(u) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) a(u, n)$$

Inverse transform

$$x(n) = \sum_{u=0}^{N-1} y(u) b(u, n)$$

basis

$$\begin{aligned} a(u, n) &= e^{-j2\pi \frac{un}{N}} \\ &= \cos(2\pi \frac{un}{N}) - j \sin(2\pi \frac{un}{N}) \end{aligned}$$



# 1-D DFT in matrix notations

$$y(u) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) a(u, n)$$

$$a(u, n) = e^{-j2\pi \frac{un}{N}}$$

$$= \cos(2\pi \frac{un}{N}) - j \sin(2\pi \frac{un}{N})$$

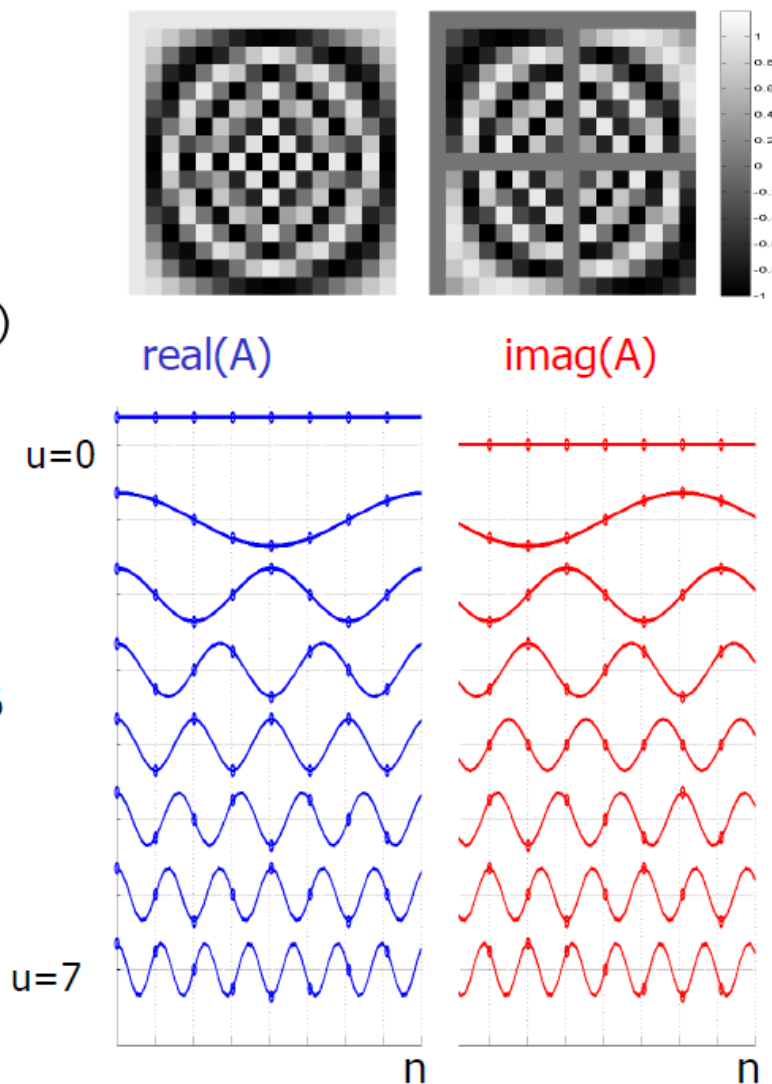
$$u = 0, 1, \dots, N-1$$



$$y = Ax$$

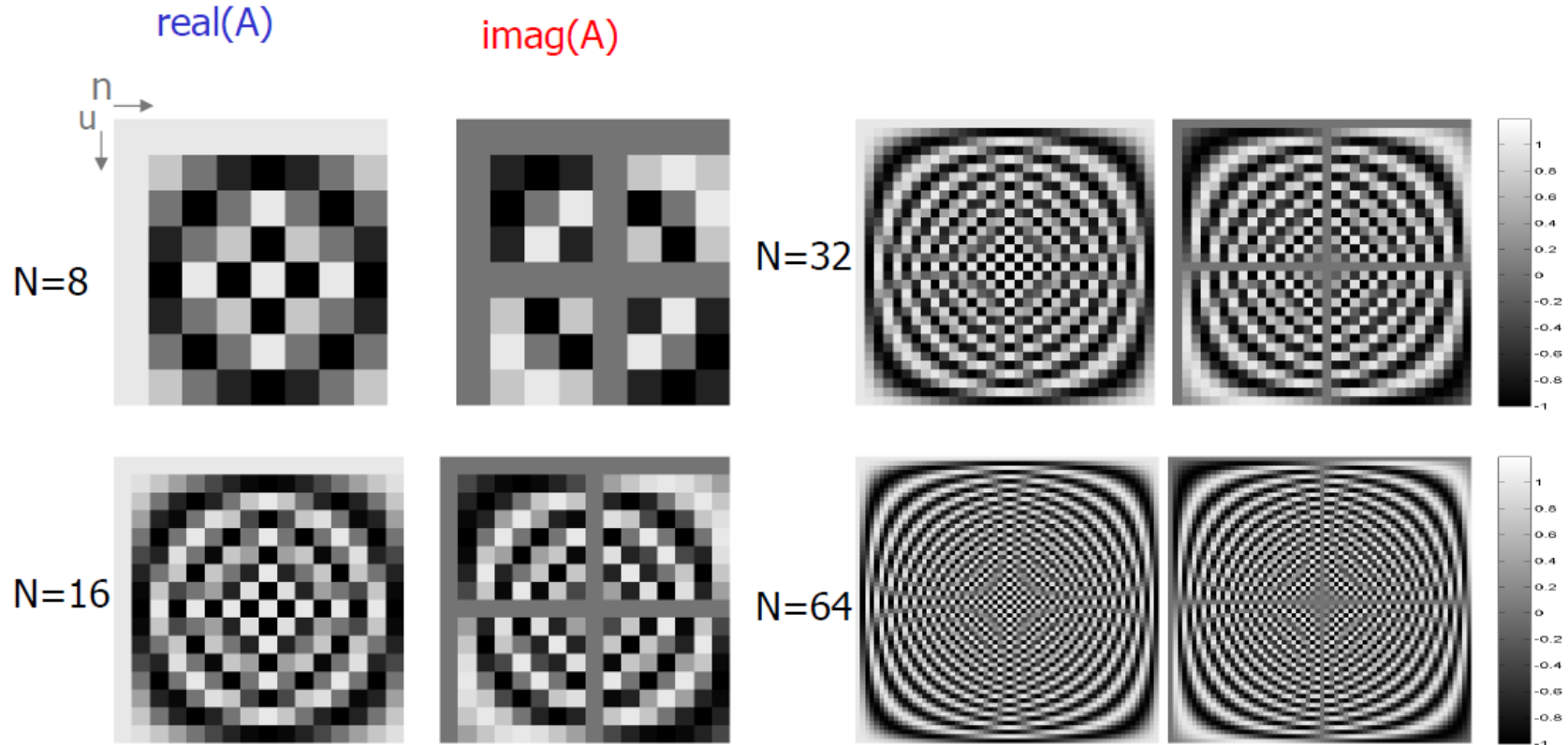
$$x = A^{-1}y$$

N=16



# 1-D DFT of different lengths

$$\begin{aligned}
 y &= Ax & a(u, n) &= e^{-j2\pi\frac{un}{N}} & u &= 0, 1, \dots, N-1 \\
 x &= A^{-1}y & &= \cos(2\pi\frac{un}{N}) - j\sin(2\pi\frac{un}{N}) & &
 \end{aligned}$$



# 1-D DFT Example

This example demonstrates how to apply the DFT to a sequence of length  $N = 4$  and the input vector

$$\mathbf{x} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 - i \\ -i \\ -1 + 2i \end{pmatrix}.$$

Calculating the DFT of  $\mathbf{x}$  using [Eq.1](#)

$$X_0 = e^{-i2\pi 0 \cdot 0/4} \cdot 1 + e^{-i2\pi 0 \cdot 1/4} \cdot (2 - i) + e^{-i2\pi 0 \cdot 2/4} \cdot (-i) + e^{-i2\pi 0 \cdot 3/4} \cdot (-1 + 2i) = 2$$

$$X_1 = e^{-i2\pi 1 \cdot 0/4} \cdot 1 + e^{-i2\pi 1 \cdot 1/4} \cdot (2 - i) + e^{-i2\pi 1 \cdot 2/4} \cdot (-i) + e^{-i2\pi 1 \cdot 3/4} \cdot (-1 + 2i) = -2 - 2i$$

$$X_2 = e^{-i2\pi 2 \cdot 0/4} \cdot 1 + e^{-i2\pi 2 \cdot 1/4} \cdot (2 - i) + e^{-i2\pi 2 \cdot 2/4} \cdot (-i) + e^{-i2\pi 2 \cdot 3/4} \cdot (-1 + 2i) = -2i$$

$$X_3 = e^{-i2\pi 3 \cdot 0/4} \cdot 1 + e^{-i2\pi 3 \cdot 1/4} \cdot (2 - i) + e^{-i2\pi 3 \cdot 2/4} \cdot (-i) + e^{-i2\pi 3 \cdot 3/4} \cdot (-1 + 2i) = 4 + 4i$$

$$\text{results in } \mathbf{X} = \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 - 2i \\ -2i \\ 4 + 4i \end{pmatrix}.$$



# From 1D to 2D

1D

$$x(n)$$

$$a(u, n)$$

$$e^{-j2\pi\frac{un}{N}}$$

$$y(u) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x(n) a(u, n)$$

$$y = Ax$$

2D

$$x(m, n)$$

$$a(u, v, m, n)$$

$$e^{-j2\pi(\frac{um}{N} + \frac{vn}{N})}$$

$$y(u, v) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x(m, n) a(u, v, m, n)$$

?

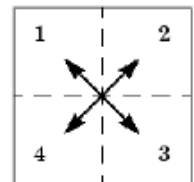
# Computing 2D-DFT

$$\text{DFT} \quad y(u, v) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x(m, n) e^{\frac{-j2\pi um}{M}} e^{\frac{-j2\pi vn}{N}}$$

$$\text{IDFT} \quad x(m, n) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} y(u, v) e^{\frac{j2\pi um}{M}} e^{\frac{j2\pi vn}{N}}$$

- Discrete, 2-D Fourier & inverse Fourier transforms are implemented in `fft2` and `ifft2`, respectively
- `fftshift`: Move origin (DC component) to image center for display
- Example:

```
>> I = imread('test.png'); % Load grayscale image
>> F = fftshift(fft2(I)); % Shifted transform
>> imshow(log(abs(F)), []); % Show log magnitude
>> imshow(angle(F), []); % Show phase angle
```



# 2D DFT from 1D DFT

The 2D Discrete Fourier Transform is given by the equation:

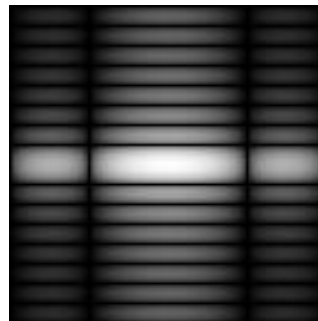
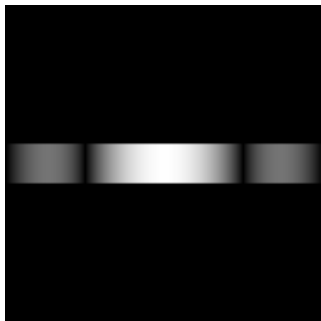
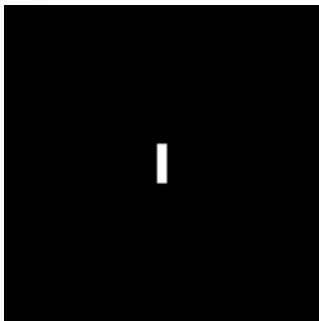
$$F(u, v) = \frac{1}{NM} \sum_{x=0}^N \sum_{y=0}^M f(x, y) e^{-j2\pi(\frac{ux}{N} + \frac{vy}{M})}$$

Which can be written as:

$$F(u, v) = \frac{1}{M} \sum_{y=0}^M \frac{1}{N} \sum_{x=0}^N f(x, y) e^{-j2\pi\frac{ux}{N}} e^{-j2\pi\frac{vy}{M}}$$

Since the kernel is separable, resulting in:

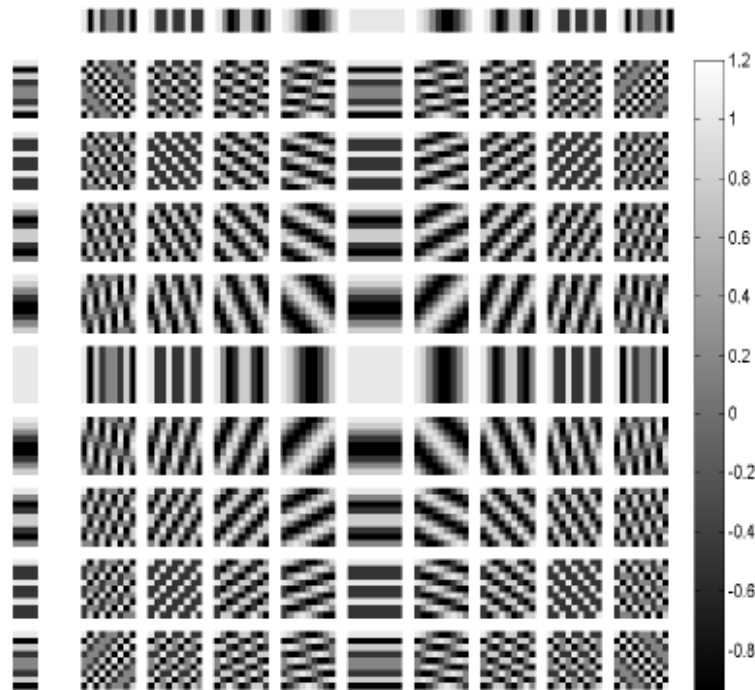
$$F(u, v) = \frac{1}{M} \sum_{y=0}^M F(u, y) e^{-j2\pi\frac{vy}{M}}$$



Left:  $f(x, y)$ , ---- Center:  $F(u, y)$ , --- Right:  $F(u, v)$

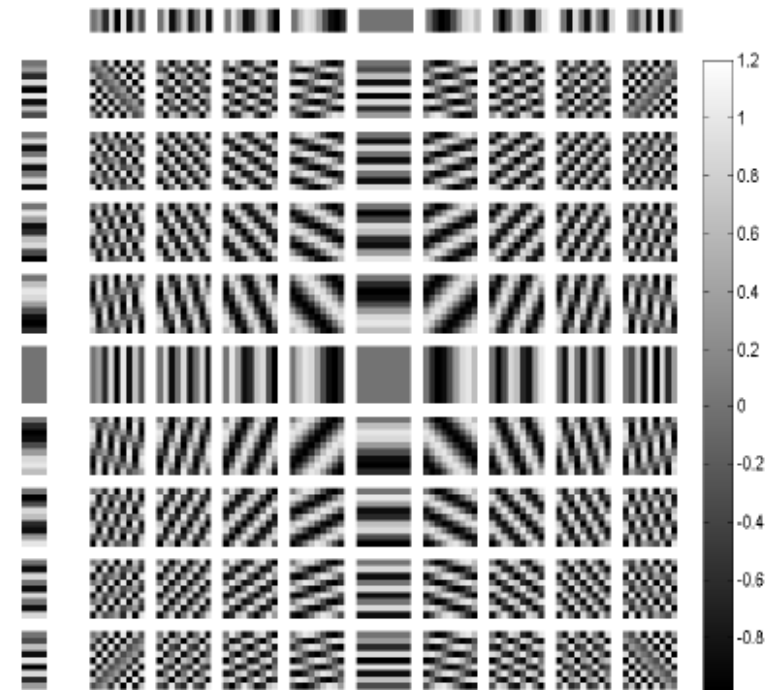
# 2-D Fourier basis

real  $e^{-j2\pi(\frac{um}{N} + \frac{vn}{N})}$



$\uparrow$   
real( $e^{-j2\pi\frac{um}{M}}$ )

imag  $e^{-j2\pi(\frac{um}{N} + \frac{vn}{N})}$



$\uparrow$   
imag( $e^{-j2\pi\frac{um}{M}}$ )

# 2-D FT illustrated

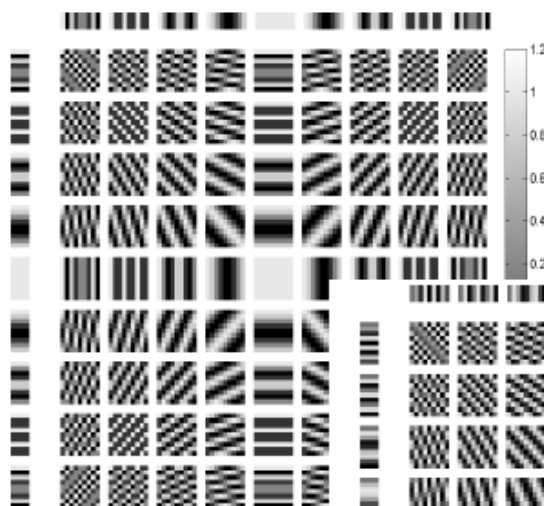
real-valued

$x(m, n)$

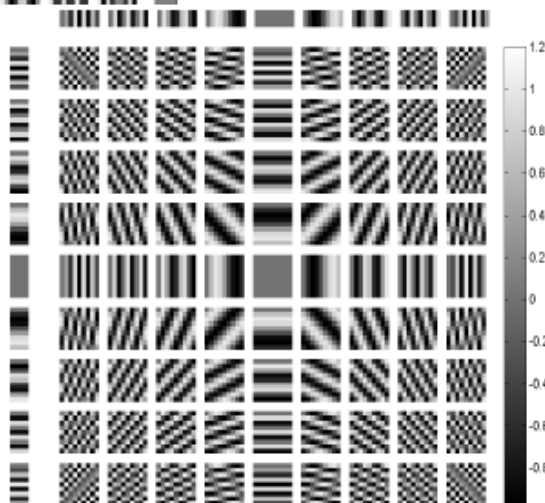


$$a(u, v, m, n) = e^{-j2\pi(\frac{um}{N} + \frac{vn}{N})}$$

real

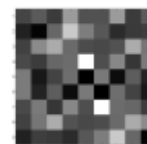


imag



→  $y(u, v)$

$real(y(u, v))$



$imag(y(u, v))$





# Notes about 2D-DFT

- Output of the Fourier transform is a complex number
  - Decompose the complex number as the magnitude and phase components
- In Matlab:  $u = \text{real}(z)$ ,  $v = \text{imag}(z)$ ,  $r = \text{abs}(z)$ , and  $\text{theta} = \text{angle}(z)$

Some useful FT pairs:

*Impulse*  $\delta(x, y) \Leftrightarrow 1$

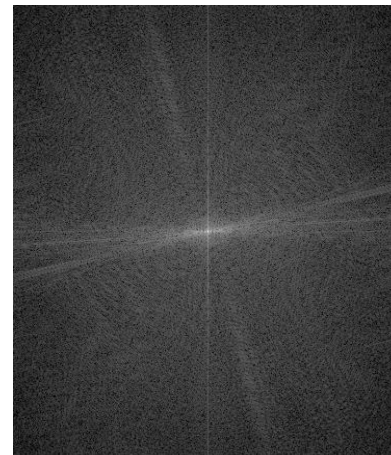
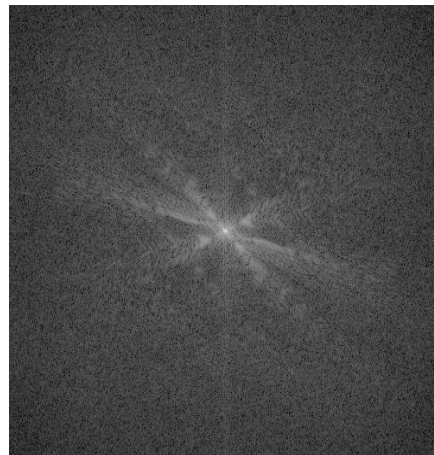
*Gaussian*  $A\sqrt{2\pi}\sigma e^{-2\pi^2\sigma^2(x^2+y^2)} \Leftrightarrow Ae^{-(u^2+v^2)/2\sigma^2}$

*Rectangle*  $\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$

*Cosine*  $\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$   
 $\frac{1}{2} [\delta(u + u_0, v + v_0) + \delta(u - u_0, v - v_0)]$

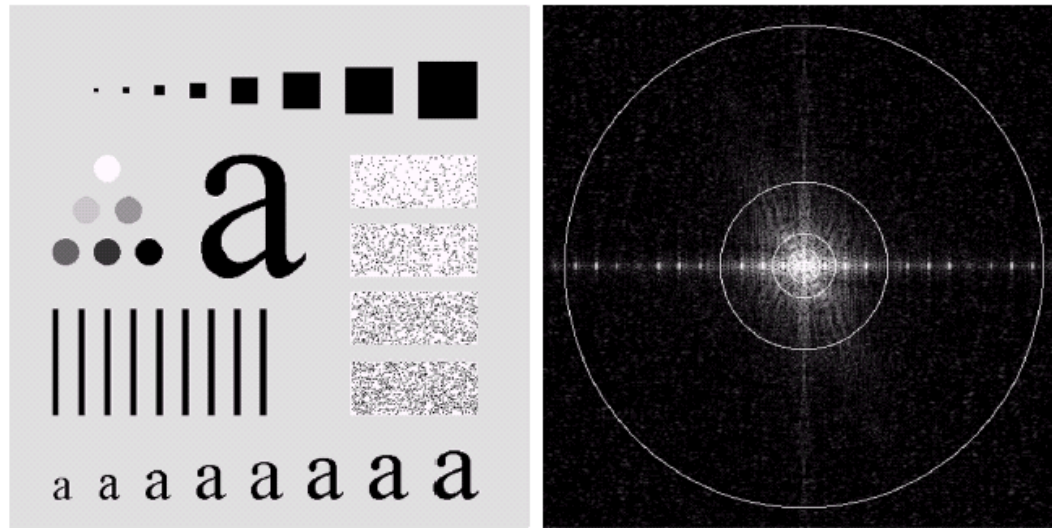
*Sine*  $\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$   
 $j \frac{1}{2} [\delta(u + u_0, v + v_0) - \delta(u - u_0, v - v_0)]$

<sup>†</sup> Assumes that functions have been extended by zero padding.



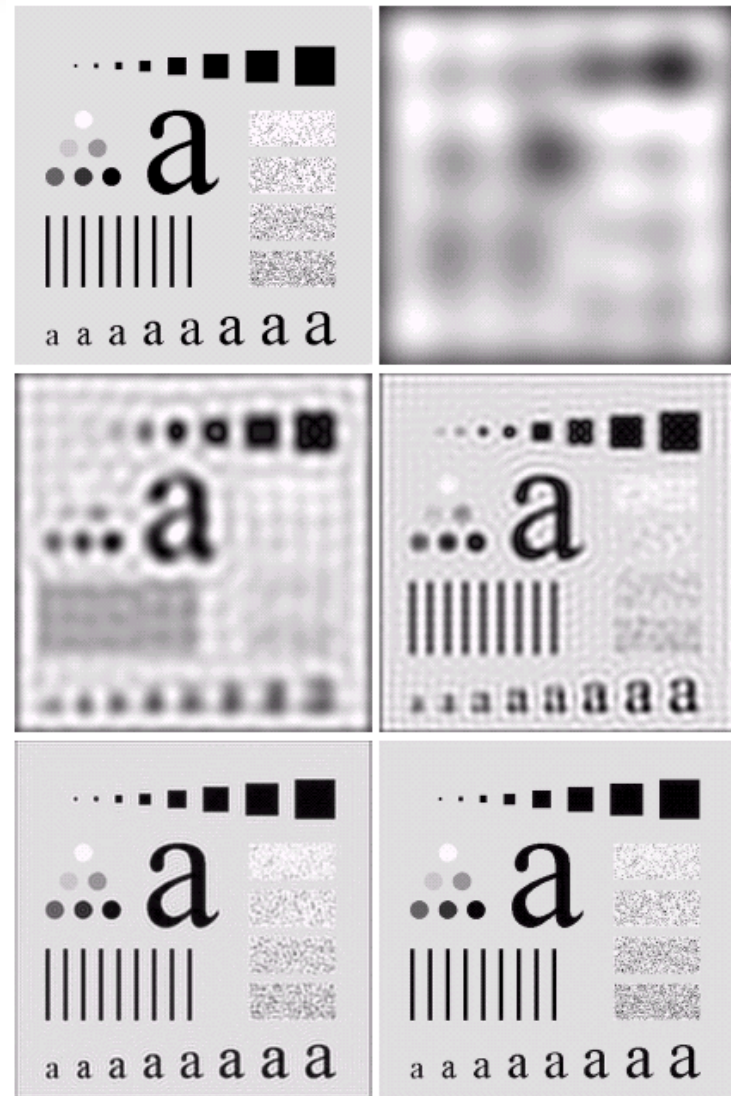


# Observation 1: compacting energy



a b

**FIGURE 4.11** (a) An image of size  $500 \times 500$  pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.



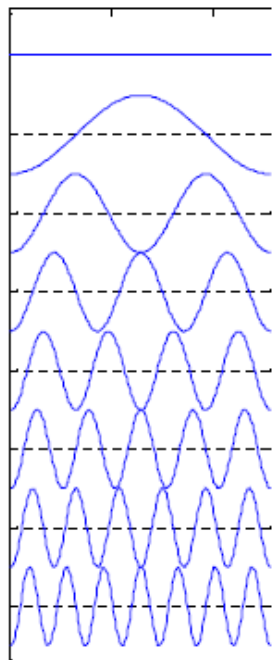
a b  
c d  
e f

**FIGURE 4.12** (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

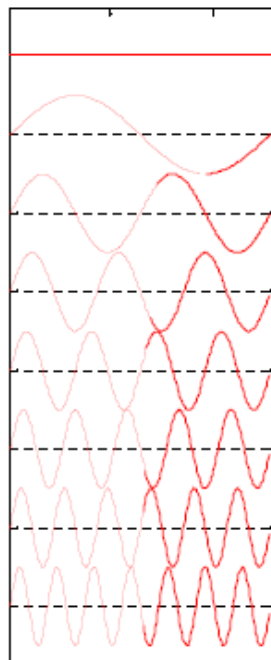
# The Phase of DFT

$$\begin{aligned}a(u, n) &= e^{-j2\pi\frac{un}{N}} \\ &= \cos(2\pi\frac{un}{N}) - j\sin(2\pi\frac{un}{N})\end{aligned}$$

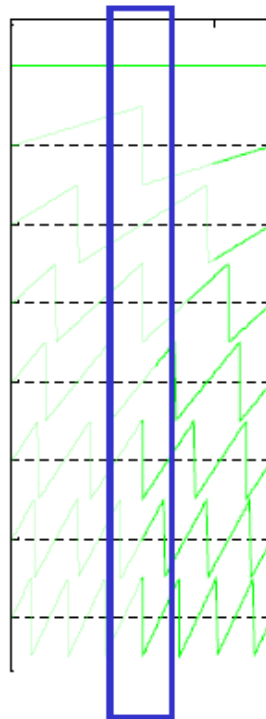
real(A)



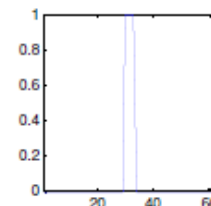
imag(A)



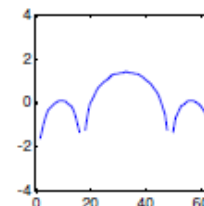
angle(A)



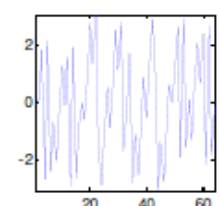
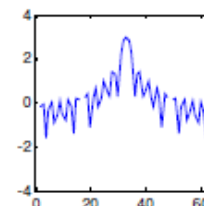
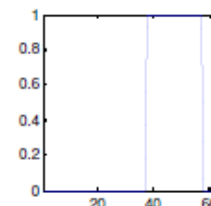
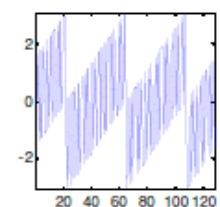
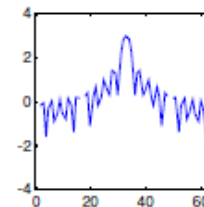
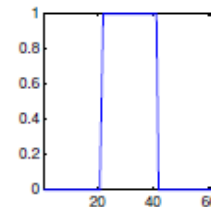
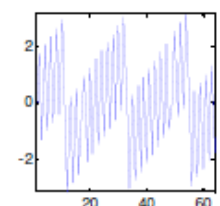
Step function



abs(FFT(.))



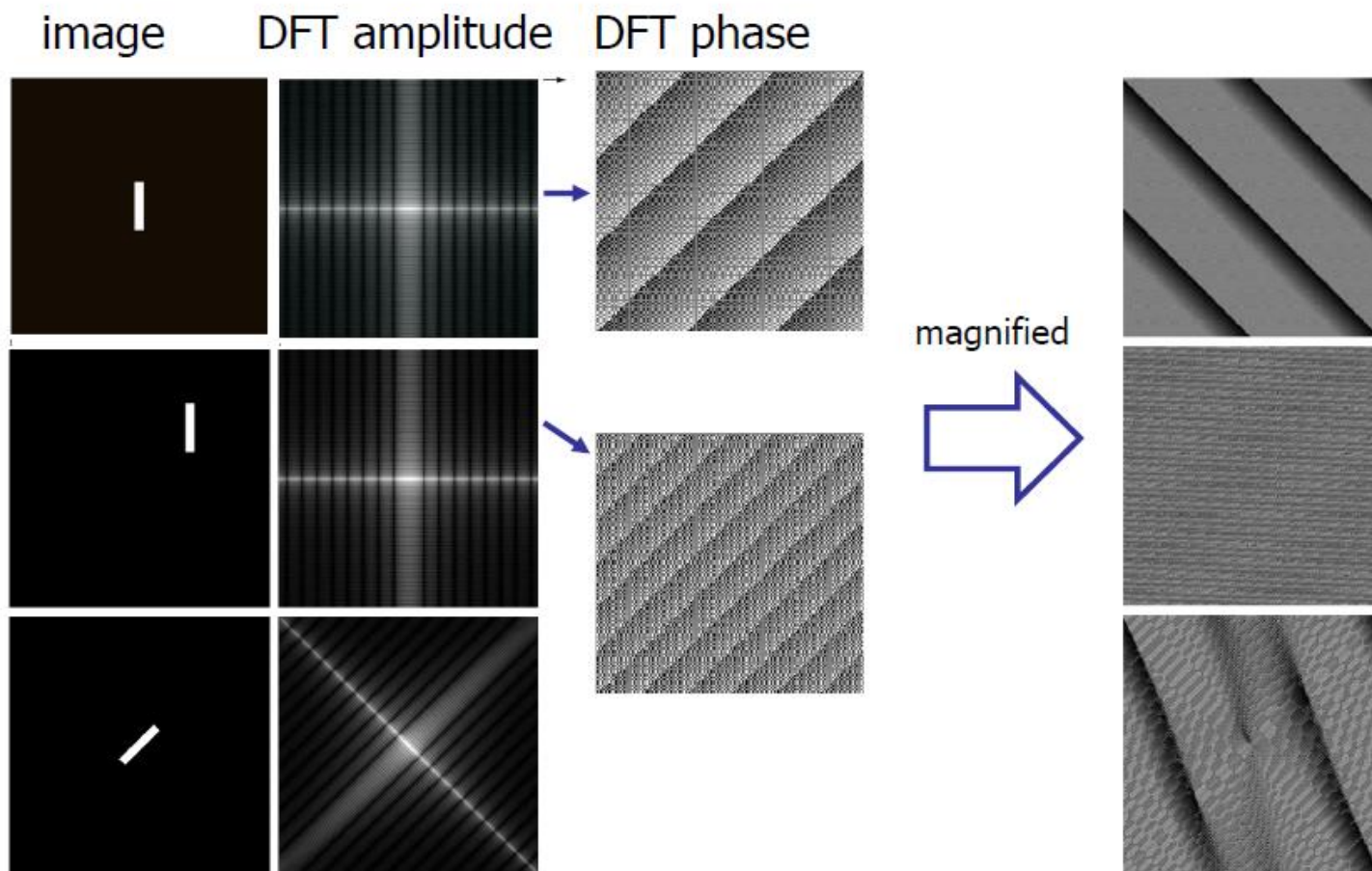
angle(FFT(.))





# Intuition of the FFT phase

- Amplitude: relative prominence of sinusoids
- Phase: relative displacement of sinusoids



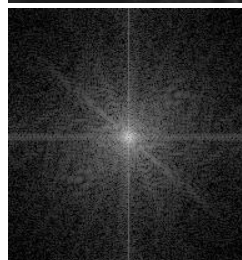


# Another example: amplitude vs. phase

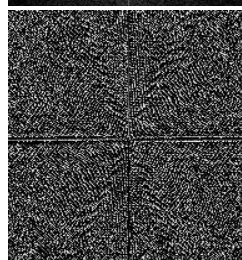
A = “Aron”  
 $FA = \text{fft2}(A)$



$\log(\text{abs}(FA))$



$\text{angle}(FA)$



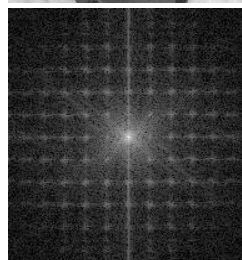
$\text{ifft2}(\text{abs}(FA), \text{angle}(FP))$



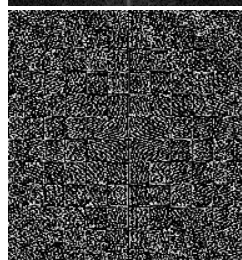
P = “Phyllis”  
 $FP = \text{fft2}(P)$



$\log(\text{abs}(FP))$



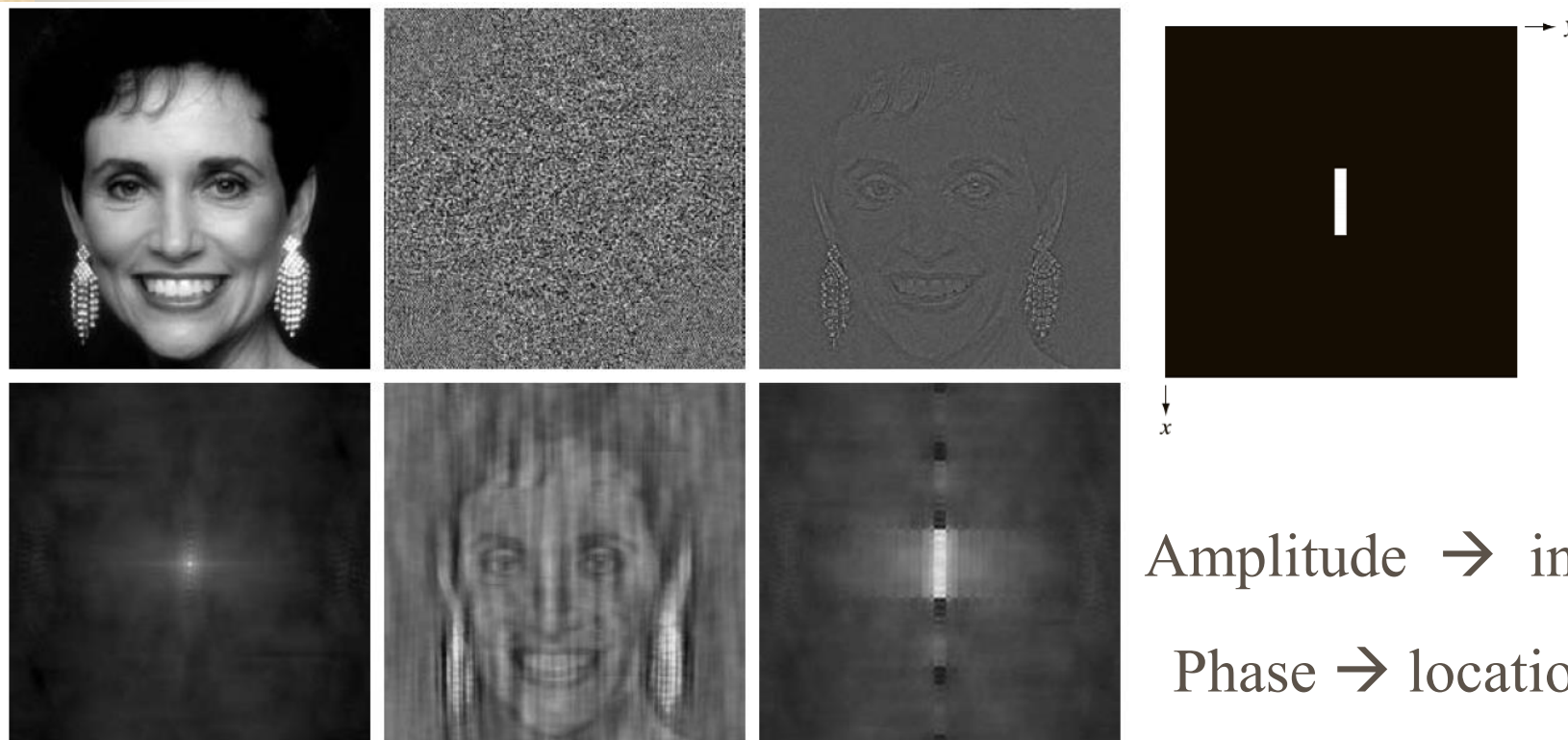
$\text{angle}(FP)$



$\text{ifft2}(\text{abs}(FP), \text{angle}(FA))$



## Observation 2: amplitude vs. phase



Amplitude  $\rightarrow$  intensity

Phase  $\rightarrow$  location

|   |   |   |
|---|---|---|
| a | b | c |
| d | e | f |

**FIGURE 4.27** (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.



# Summary of Observations

- Magnitude → how strong the pattern is
- Phase → where the pattern is located
- Low frequencies → smooth regions, illumination
- High frequencies → edges, fine details, noise
- Very high frequencies → sharp transitions



# Fast implementation of 2-D DFT (FFT)

- 2 Dimensional DFT is separable

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(m, n) e^{-\frac{2\pi j u m}{M}} e^{-\frac{2\pi j v n}{N}}$$

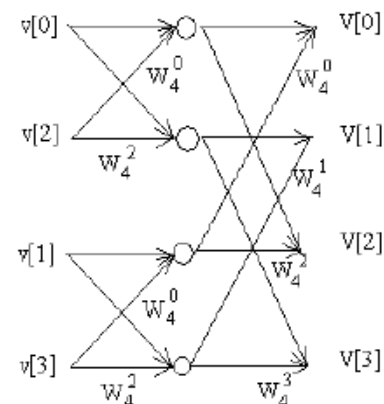
$$= \frac{1}{M} \sum_{x=0}^{M-1} e^{-\frac{2\pi j u m}{M}} \cdot \frac{1}{N} \sum_{y=0}^{N-1} f(m, n) e^{-\frac{2\pi j v n}{N}}$$

1-D DFT  
of  $f(m, n)$   
w.r.t  $n$

$$= \frac{1}{M} \sum_{x=0}^{M-1} e^{-\frac{2\pi j u m}{M}} F(m, v)$$

1-D DFT  
of  $F(m, v)$   
w.r.t  $m$

- 1D FFT:  $O(N \log_2 N)$
- 2D DFT naïve implementation:  $O(N^4)$
- 2D DFT as 1D FFT for each row and then for each column:  $O(N^2 \log_2 N)$



More details: [http://en.wikipedia.org/wiki/Fast\\_Fourier\\_transform](http://en.wikipedia.org/wiki/Fast_Fourier_transform)



# Implement IDFT as DFT

$$\text{DFT2} \quad F(u, v) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) e^{-j2\pi(\frac{um}{M} + \frac{vn}{N})}$$

$$\text{IDFT2} \quad f(m, n) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{um}{M} + \frac{vn}{N})}$$

$$\begin{aligned} \Rightarrow f^*(m, n) &= \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(\frac{um}{M} + \frac{vn}{N})} \\ &= (MN) \cdot \text{DFT2}[F^*(u, v)] \end{aligned}$$

# Properties of 2D-DFT - 1

| Name                                                      | Expression(s)                                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1) Discrete Fourier transform (DFT) of $f(x, y)$          | $F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M+vy/N)}$                     |
| 2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$ | $f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M+vy/N)}$         |
| 3) Polar representation                                   | $F(u, v) =  F(u, v)  e^{j\phi(u, v)}$                                                           |
| 4) Spectrum                                               | $ F(u, v)  = [R^2(u, v) + I^2(u, v)]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$      |
| 5) Phase angle                                            | $\phi(u, v) = \tan^{-1} \left[ \frac{I(u, v)}{R(u, v)} \right]$                                 |
| 6) Power spectrum                                         | $P(u, v) =  F(u, v) ^2$                                                                         |
| 7) Average value                                          | $\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$ |

(Continued)

**TABLE 4.2**

Summary of DFT definitions and corresponding expressions.

# Properties of 2D-DFT - 2

| Name                                                                             | Expression(s)                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8) Periodicity ( $k_1$ and $k_2$ are integers)                                   | $F(u, v) = F(u + k_1M, v) = F(u, v + k_2N)$ $= F(u + k_1M, v + k_2N)$ $f(x, y) = f(x + k_1M, y) = f(x, y + k_2N)$ $= f(x + k_1M, y + k_2N)$                                                                                                                                                                                                                                             |
| 9) Convolution                                                                   | $f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$                                                                                                                                                                                                                                                                                                      |
| 10) Correlation                                                                  | $f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n)h(x + m, y + n)$                                                                                                                                                                                                                                                                                                    |
| 11) Separability                                                                 | The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.                                                                                                                                                                                               |
| 12) Obtaining the inverse Fourier transform using a forward transform algorithm. | $MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v)e^{-j2\pi(ux/M+vy/N)}$ <p>This equation indicates that inputting <math>F^*(u, v)</math> into an algorithm that computes the forward transform (right side of above equation) yields <math>MNf^*(x, y)</math>. Taking the complex conjugate and dividing by <math>MN</math> gives the desired inverse. See Section 4.11.2.</p> |

**TABLE 4.2**

Summary of DFT definitions and corresponding expressions.

# Properties of 2D-DFT - 3

| Name                                                              | DFT Pairs                                                                                                                                                                          |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) Symmetry properties                                            | See Table 4.1                                                                                                                                                                      |
| 2) Linearity                                                      | $af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$                                                                                                                  |
| 3) Translation (general)                                          | $f(x, y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$<br>$f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M + vy_0/N)}$                                |
| 4) Translation to center of the frequency rectangle, $(M/2, N/2)$ | $f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$<br>$f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$                                                                 |
| 5) Rotation                                                       | $f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$<br>$x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$ |
| 6) Convolution theorem <sup>†</sup>                               | $f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$<br>$f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$                                                                   |

(Continued)

**TABLE 4.3**

Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the closed-form, continuous expressions.



# Properties of 2D-DFT - 4

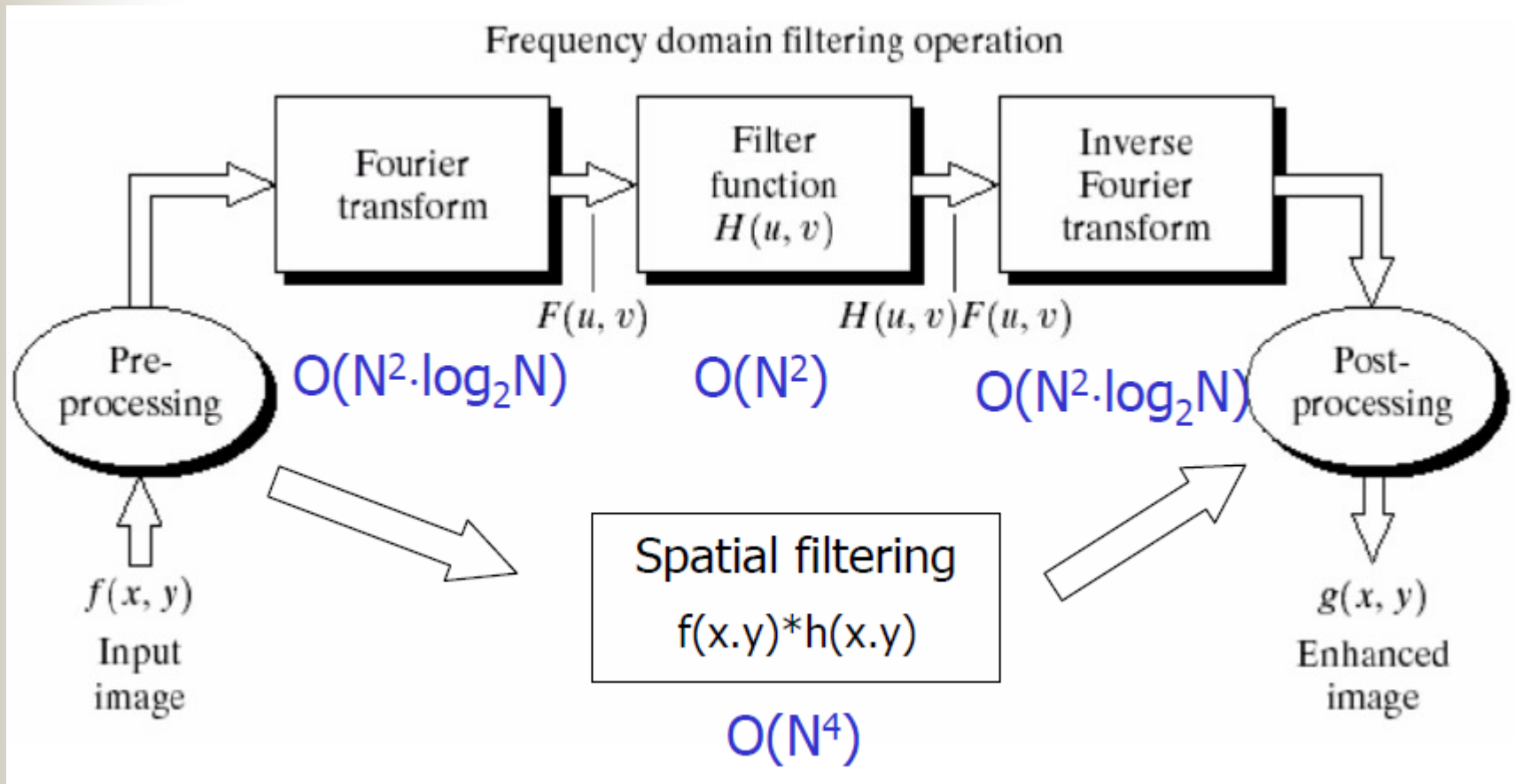
| Name                                                                                                                                                                                                                                                                                                                  | DFT Pairs                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7) Correlation theorem <sup>†</sup>                                                                                                                                                                                                                                                                                   | $f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v) H(u, v)$<br>$f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$                                                                                                                                                                                                        |
| 8) Discrete unit impulse                                                                                                                                                                                                                                                                                              | $\delta(x, y) \Leftrightarrow 1$                                                                                                                                                                                                                                                                                              |
| 9) Rectangle                                                                                                                                                                                                                                                                                                          | $\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$                                                                                                                                                                                                           |
| 10) Sine                                                                                                                                                                                                                                                                                                              | $\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$<br>$j \frac{1}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$                                                                                                                                                                                                  |
| 11) Cosine                                                                                                                                                                                                                                                                                                            | $\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$<br>$\frac{1}{2} [\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)]$                                                                                                                                                                                                    |
| <p>The following Fourier transform pairs are derivable only for continuous variables, denoted as before by <math>t</math> and <math>z</math> for spatial variables and by <math>\mu</math> and <math>\nu</math> for frequency variables. These results can be used for DFT work by sampling the continuous forms.</p> |                                                                                                                                                                                                                                                                                                                               |
| 12) Differentiation<br>(The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$ .)                                                                                                                                                                                                                     | $\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$<br>$\frac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$ |
| 13) Gaussian                                                                                                                                                                                                                                                                                                          | $A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2}$ ( $A$ is a constant)                                                                                                                                                                                                             |

**TABLE 4.3**

Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the closed-form, continuous expressions.

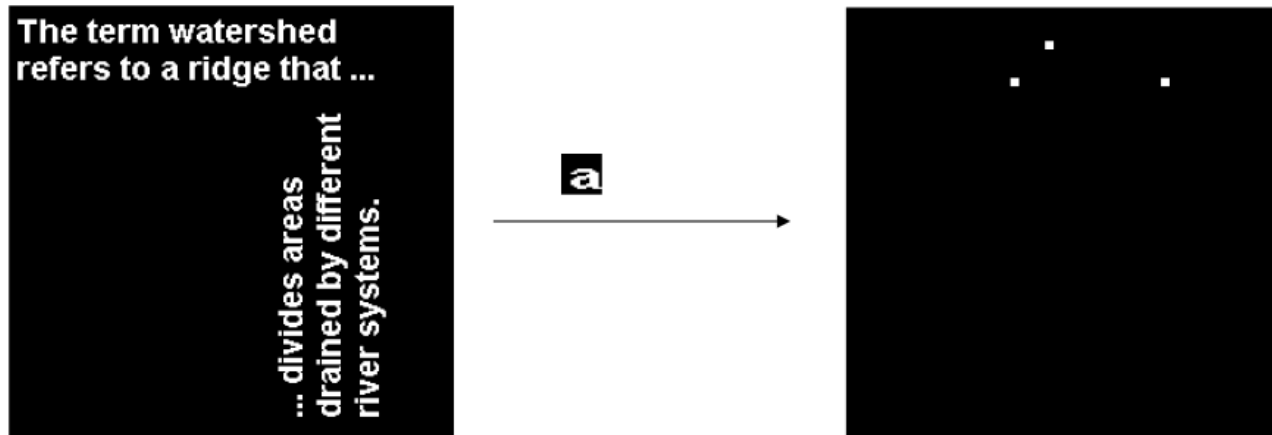
<sup>†</sup> Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

# DFT application #1: fast Convolution



# DFT application #2: feature correlation

- Find letter "a" in the following image

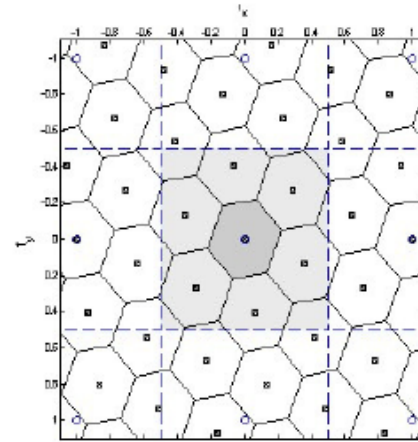
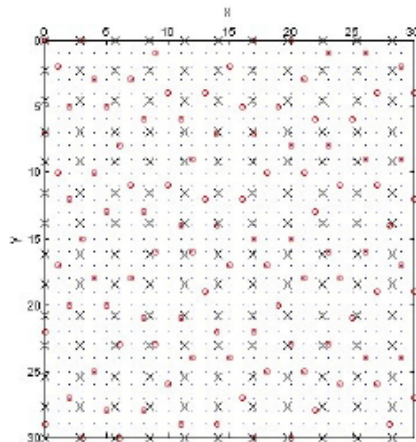


```
bw = imread('text.png'); a = imread('letter_a.png');  
% Convolution is equivalent to correlation if you rotate the  
% convolution kernel by 180deg  
C = real(ifft2(fft2(bw) .* fft2(rot90(a,2),256,256)));  
% Use a threshold that's a little less than max.  
% Display showing pixels over threshold.  
thresh = .9*max(C(:)); figure, imshow(C > thresh)
```

from Matlab image processing demos.

# DFT application #3: image filters

- A zoology of image filters
  - Smoothing / Sharpening / Others
  - Support in time/space vs. support in frequency c.f. "FIR / IIR"
  - Definition: spatial domain/frequency domain
  - Separable / Non-separable







# Convolution Property of the Fourier Transform

Let functions  $f(r, c)$  and  $g(r, c)$  have  
Fourier Transforms  $F(u, v)$  and  $G(u, v)$ .

Then,

$$\mathcal{F}\{f * g\} = F \cdot G.$$

Moreover,

$$\mathcal{F}\{f \cdot g\} = F * G.$$

$*$  = convolution

$\cdot$  =

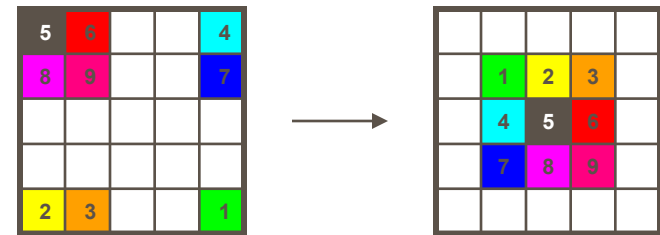
multiplication

The Fourier Transform of a convolution equals the product of the Fourier Transforms. Similarly, the Fourier Transform of a convolution is the product of the Fourier Transforms

# Matlab's `fftshift` and `ifftshift`

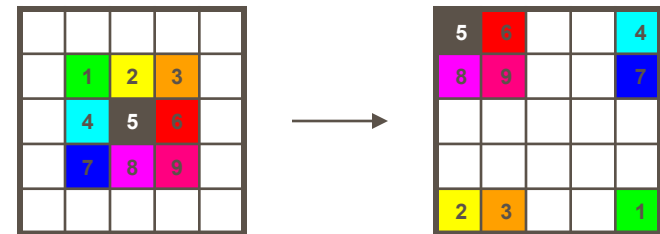
$J = \text{fftshift}(I) :$

$$I(1,1) \rightarrow J(\lfloor M/2 \rfloor + 1, \lfloor N/2 \rfloor + 1)$$



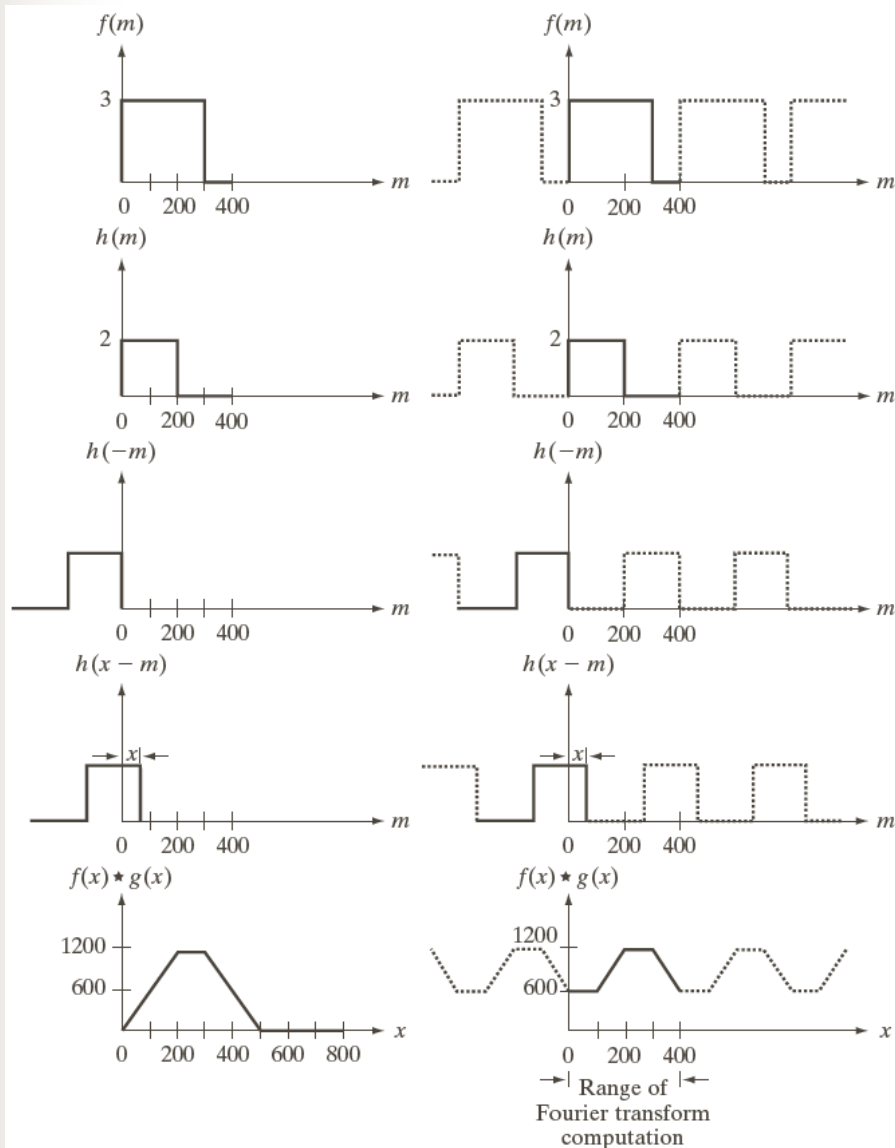
$I = \text{ifftshift}(J) :$

$$J(\lfloor M/2 \rfloor + 1, \lfloor N/2 \rfloor + 1) \rightarrow I(1,1)$$



where  $\lfloor x \rfloor = \text{floor}(x)$  = the largest integer smaller than  $x$ .

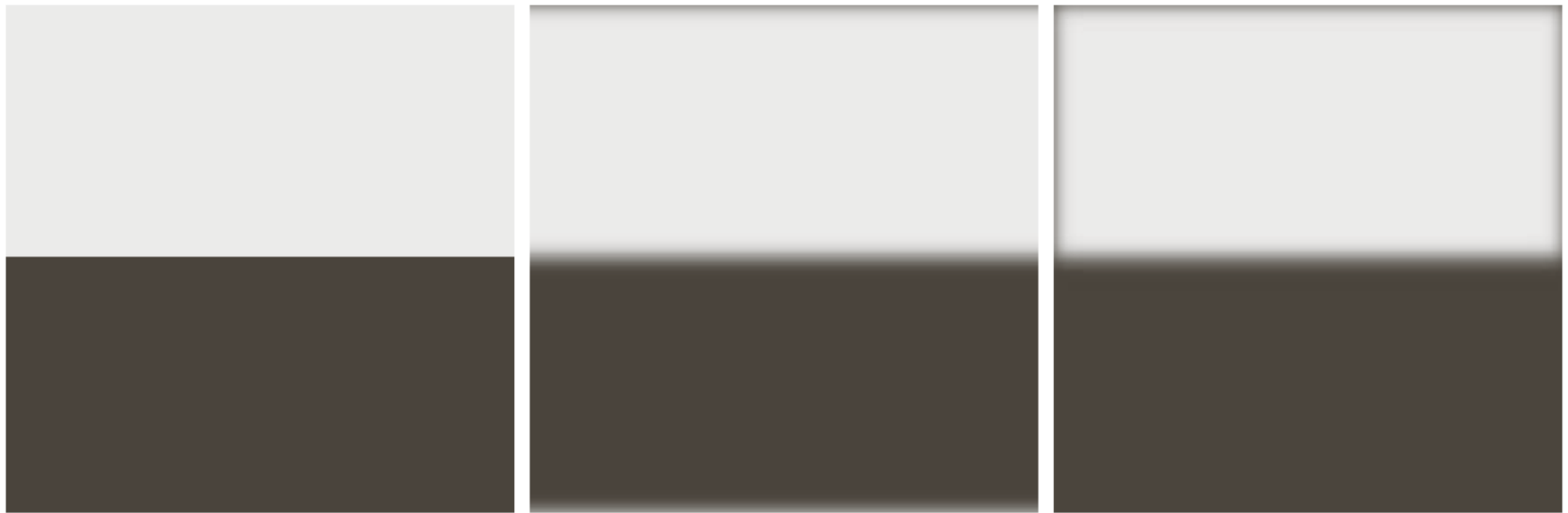
# Circular Convolution and Zero-padding



|   |   |
|---|---|
| a | f |
| b | g |
| c | h |
| d | i |
| e | j |

**FIGURE 4.28** Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

# Example of zero-padding



a b c

**FIGURE 4.32** (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with padding. Compare the light area of the vertical edges in (b) and (c).





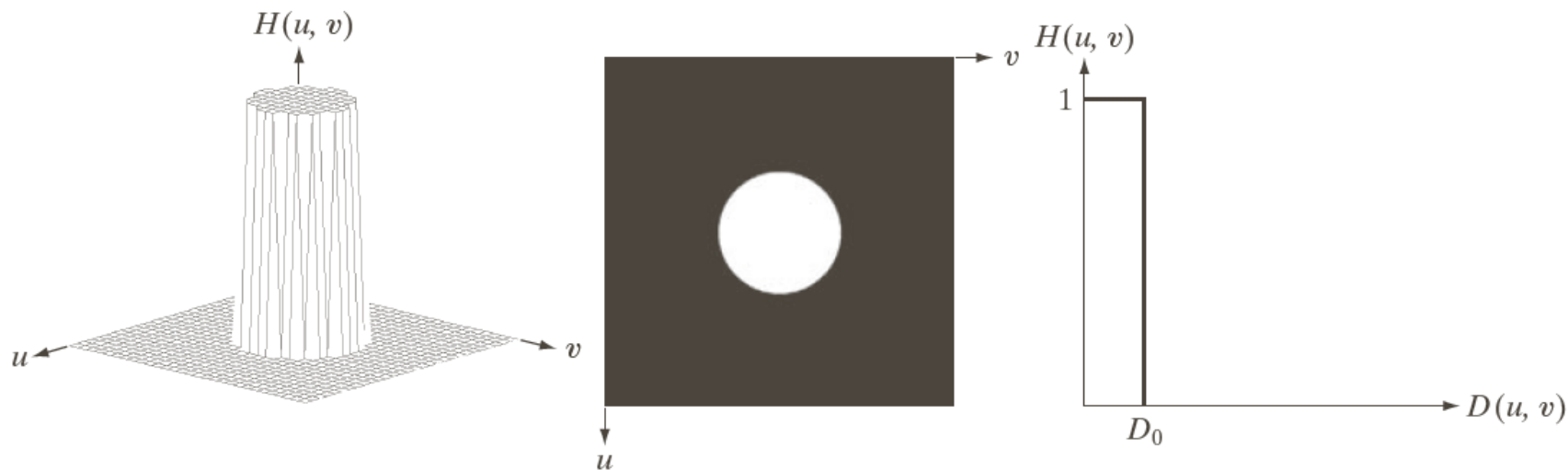
# Smoothing/ Lowpass Filtering

Smoothing results from:

- Pixel averaging in the spatial domain:
  - Each pixel in the output is a weighted average of its neighbors.
  - Is a convolution whose weight matrix sums to 1.
- Lowpass filtering in the frequency domain:
  - High frequencies are diminished or eliminated
  - Individual frequency components are multiplied by a nonincreasing function of  $\omega$  such as  $1/\omega = 1/\sqrt{u^2+v^2}$ .

The values of the output image are all non-negative.

# Ideal low-pass

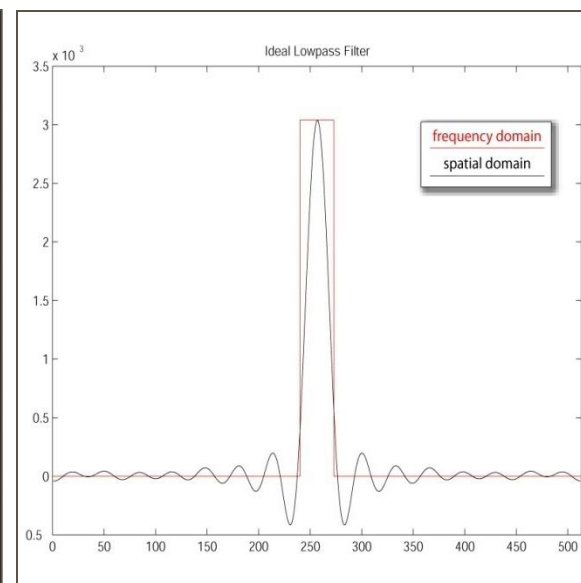
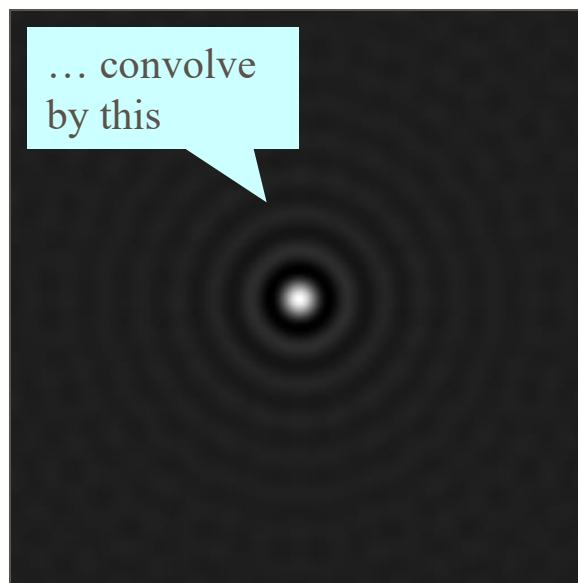
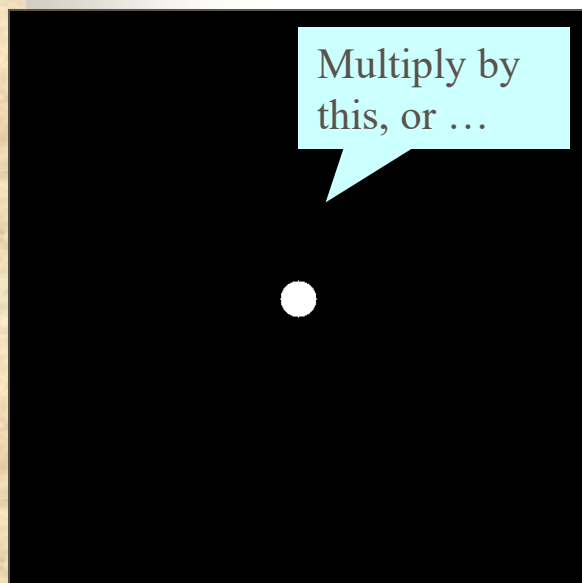


a b c

**FIGURE 4.40** (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

# Ideal Lowpass Filter

Image size: 512x512  
FD filter radius: 16



Fourier Domain Rep.

Spatial Representation

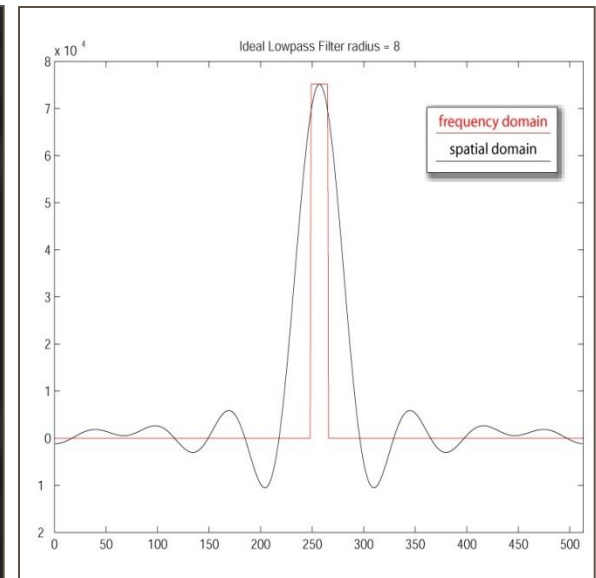
Central Profile

# Ideal Lowpass Filter

Image size: 512x512  
FD filter radius: 8

Multiply by  
this, or ...

... convolve  
by this

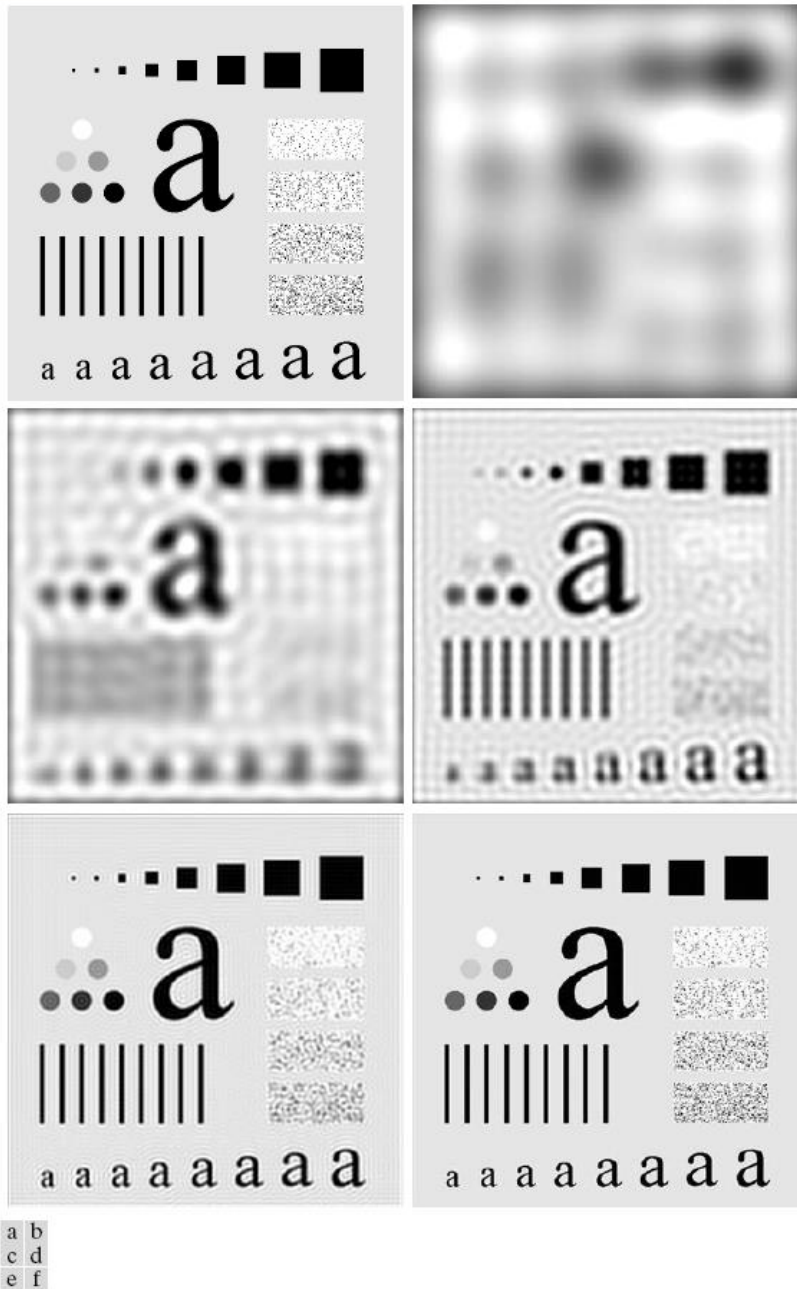


Fourier Domain Rep.

Spatial Representation

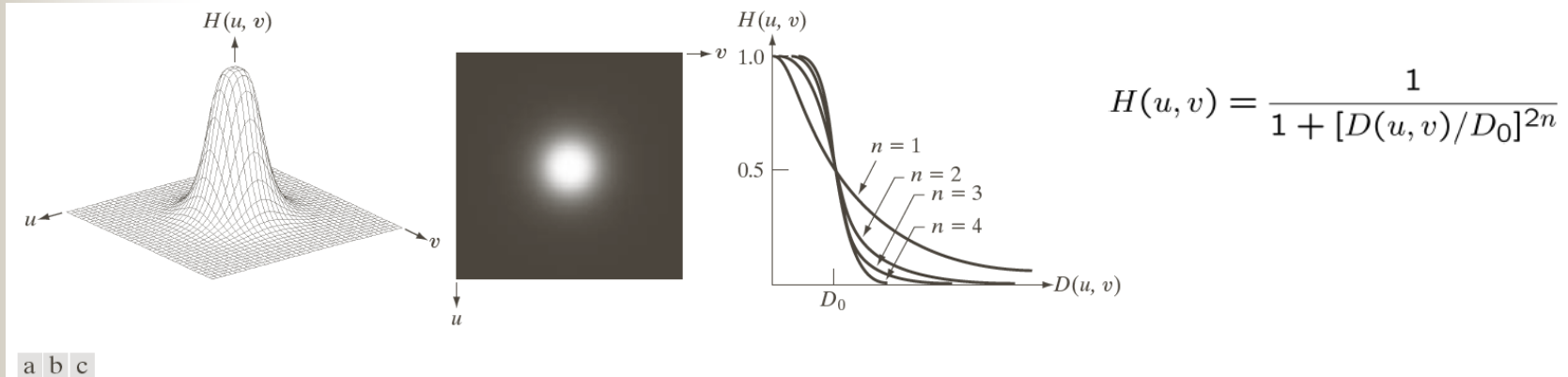
Central Profile



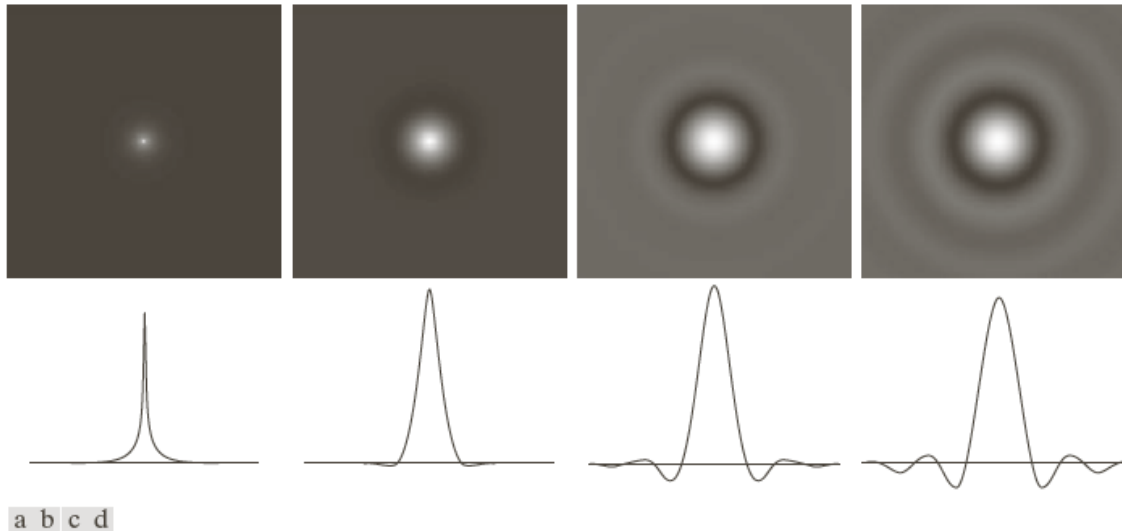


**FIGURE 4.42** (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

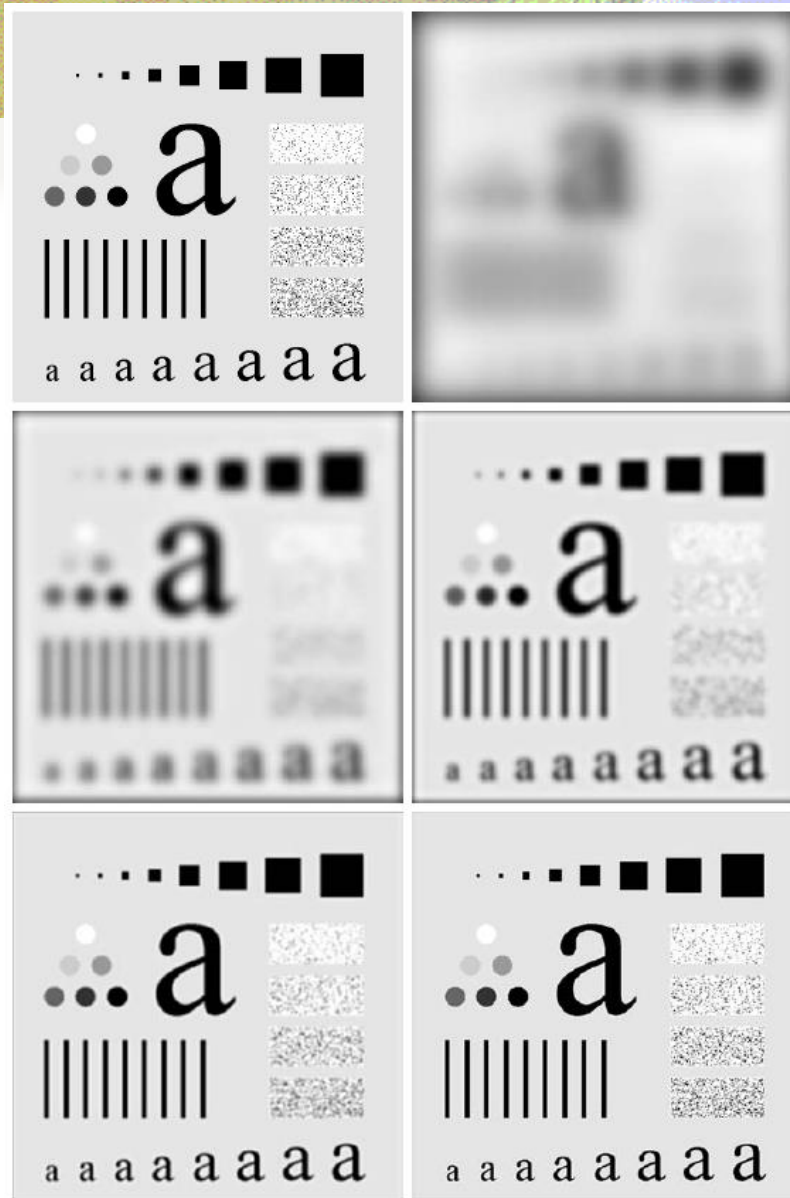
# Butterworth lowpass filters



**FIGURE 4.44** (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.



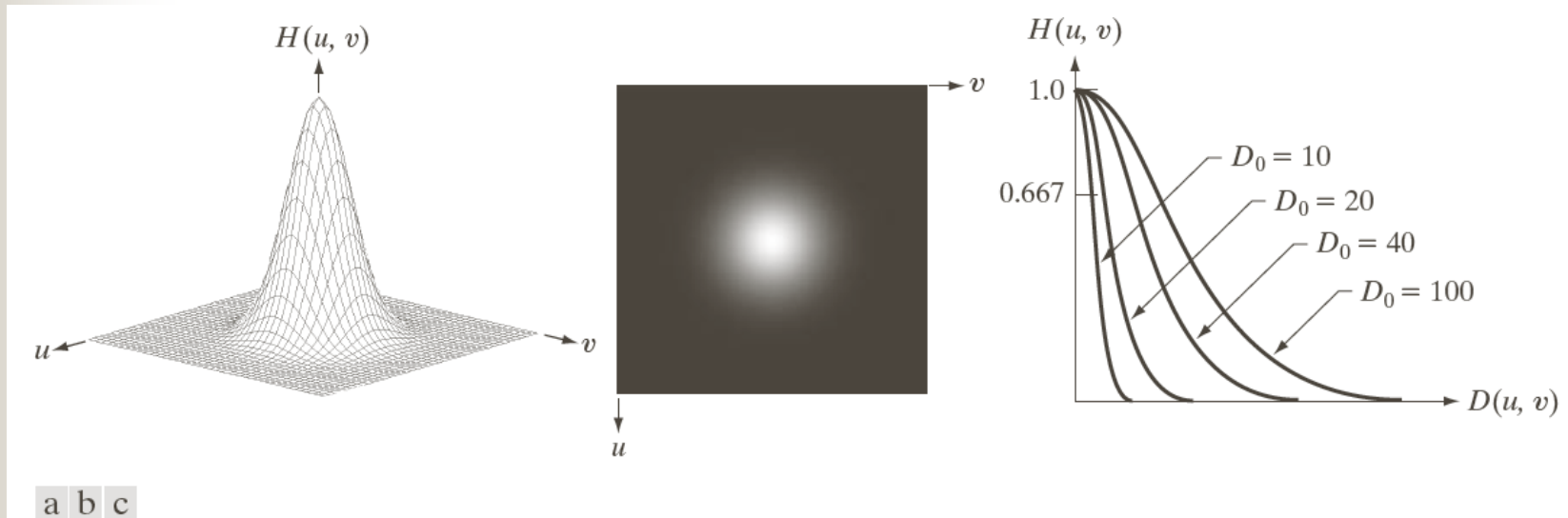
**FIGURE 4.46** (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding intensity profiles through the center of the filters (the size in all cases is  $1000 \times 1000$  and the cutoff frequency is 5). Observe how ringing increases as a function of filter order.



a b  
c d  
e f

**FIGURE 4.45** (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

# Gaussian lowpass filters



**FIGURE 4.47** (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of  $D_0$ .

$$H(u, v) = e^{-D^2(u, v)/2\sigma^2}$$

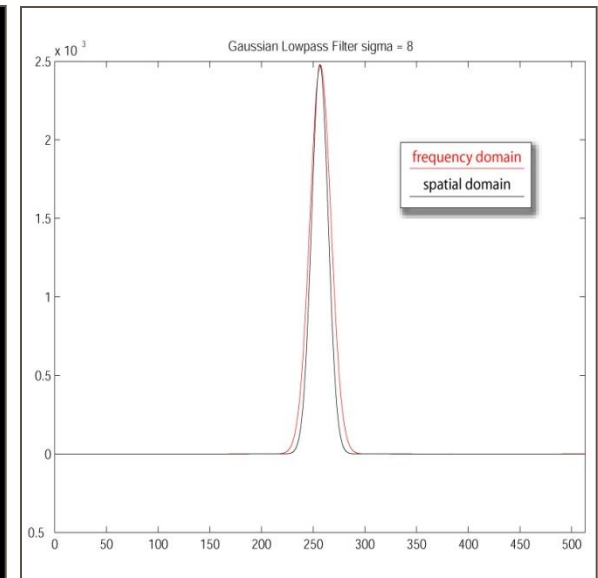


# Gaussian Lowpass Filter

Image size: 512x512  
SD filter sigma = 8

Multiply by  
this, or ...

... convolve  
by this



Fourier Domain Rep.

Spatial Representation

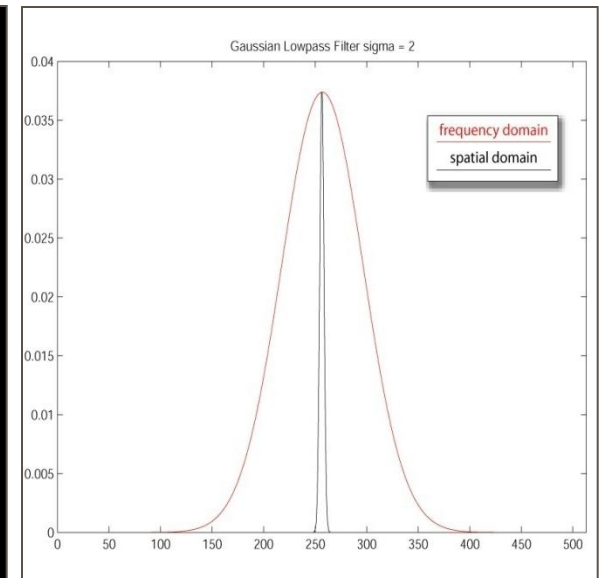
Central Profile

# Gaussian Lowpass Filter

Image size: 512x512  
SD filter sigma = 2

Multiply by  
this, or ...

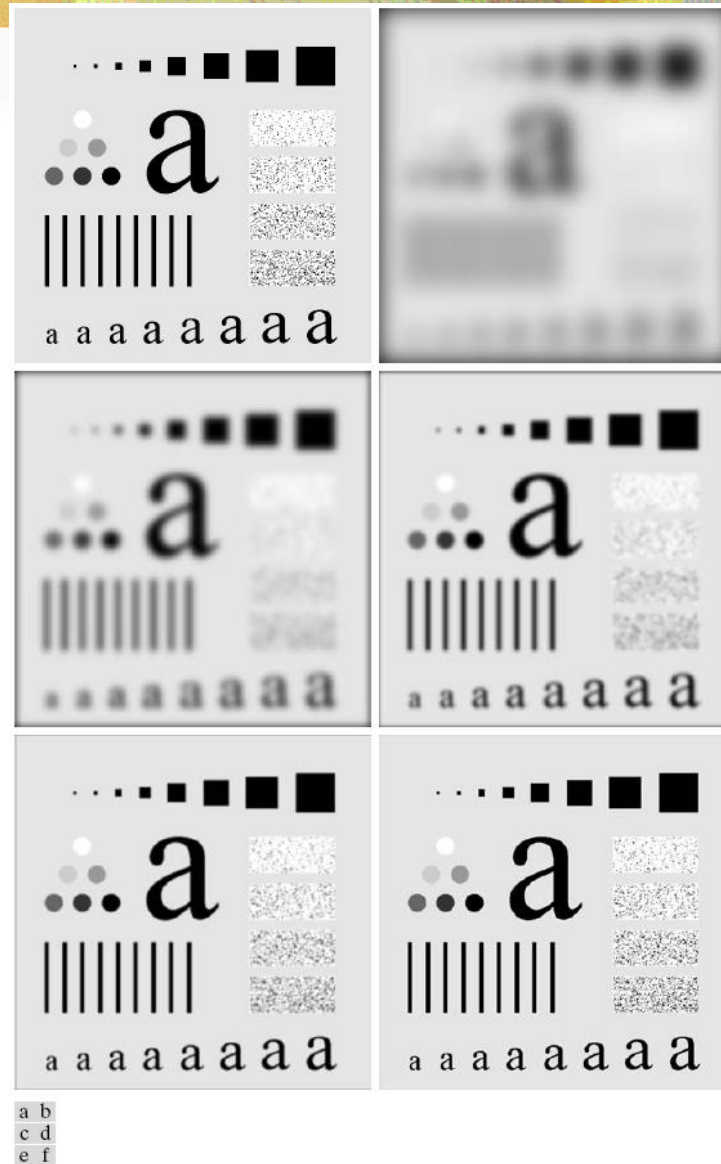
... convolve  
by this



Fourier Domain Rep.

Spatial Representation

Central Profile



**FIGURE 4.48** (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.

# Summary of lowpass filters

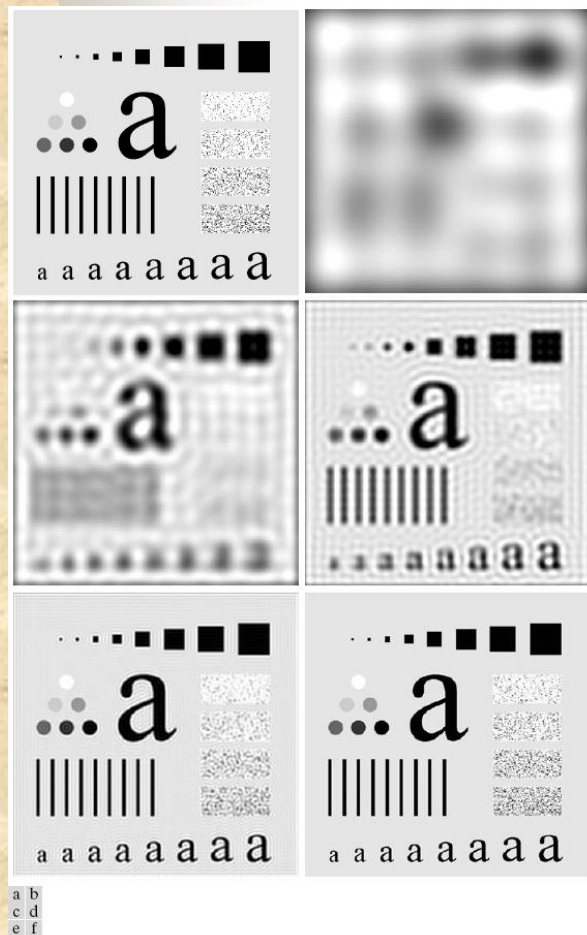
**TABLE 4.4**

Lowpass filters.  $D_0$  is the cutoff frequency and  $n$  is the order of the Butterworth filter.

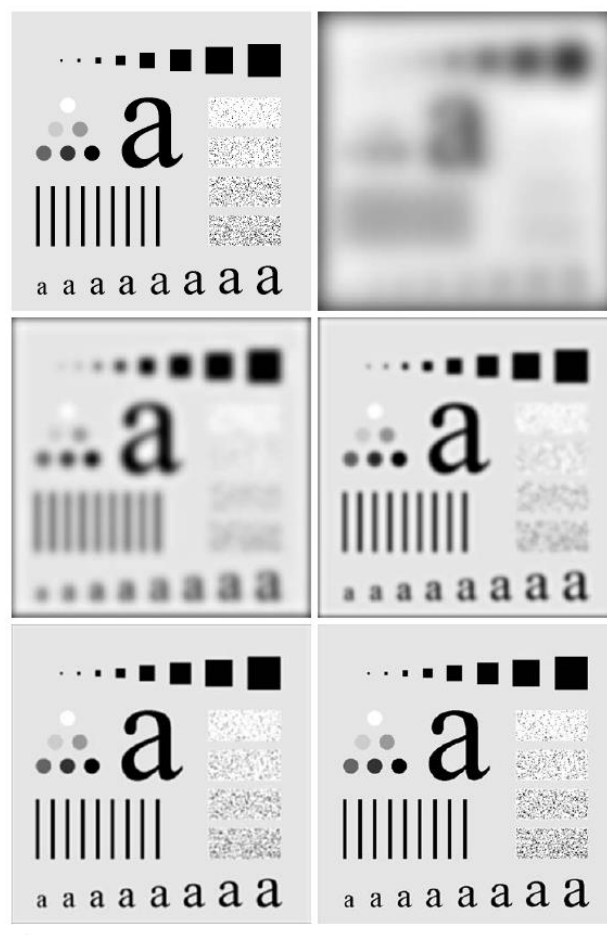
| Ideal                                                                                                 | Butterworth                                  | Gaussian                         |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------|
| $H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$ | $H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$ | $H(u, v) = e^{-D^2(u,v)/2D_0^2}$ |



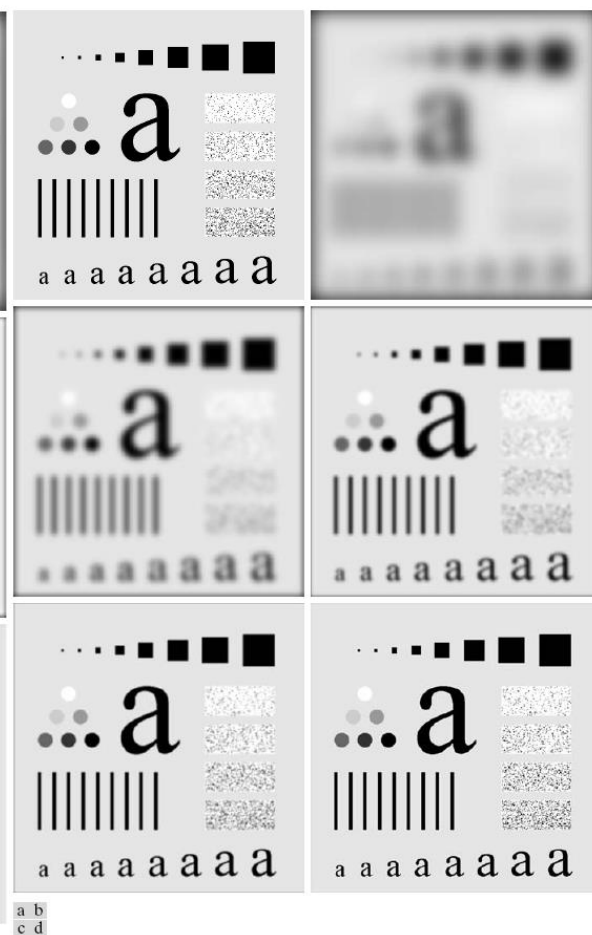
# Examples of lowpass filters



**FIGURE 4.42** (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.



**FIGURE 4.45** (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.



**FIGURE 4.48** (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs 4.42 and 4.45.

# Smoothing filter application 1

## text enhancement

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



a b

**FIGURE 4.49**  
(a) Sample text of low resolution (note broken characters in magnified view).  
(b) Result of filtering with a GLPF (broken character segments were joined).

# Smoothing filter application 2

beautify a photo



a b c

**FIGURE 4.50** (a) Original image ( $784 \times 732$  pixels). (b) Result of filtering using a GLPF with  $D_0 = 100$ . (c) Result of filtering using a GLPF with  $D_0 = 80$ . Note the reduction in fine skin lines in the magnified sections in (b) and (c).





# Sharpening: Highpass Filtering

Sharpening results from adding to the image, a copy of itself that has been:

- Pixel-differenced in the spatial domain:
  - Each pixel in the output is a difference between itself and a weighted average of its neighbors.
  - Is a convolution whose weight matrix sums to 0.
- Highpass filtered in the frequency domain:
  - High frequencies are enhanced or amplified.
  - Individual frequency components are multiplied by an increasing function of  $\omega$  such as  $\alpha\omega = \alpha\sqrt{(u^2+v^2)}$ , where  $\alpha$  is a constant.

The values of the output image positive & negative.



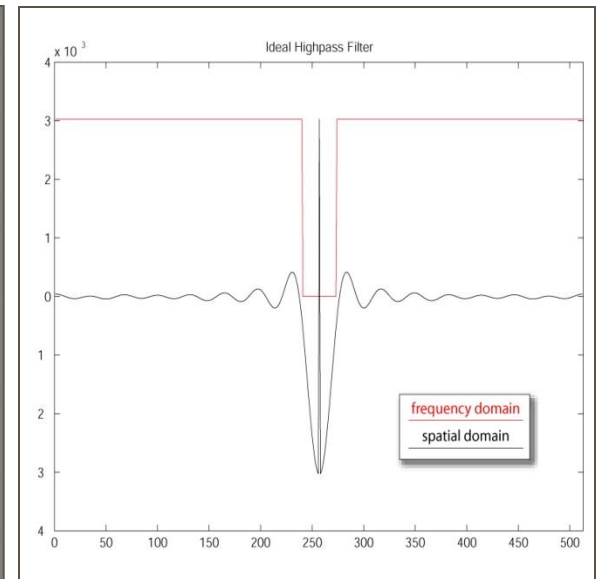
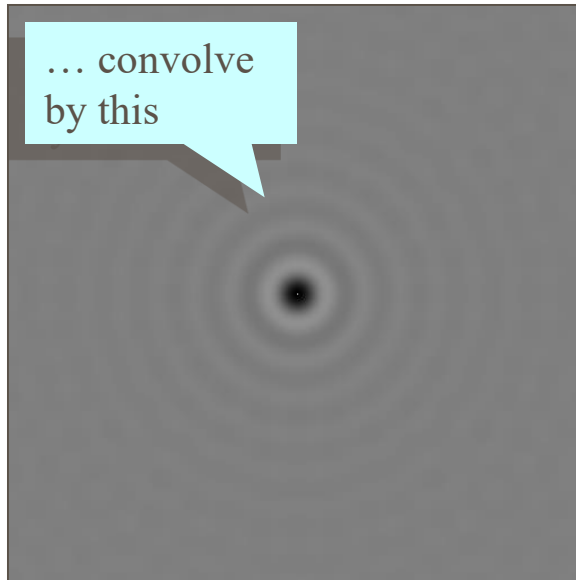
# Ideal Highpass Filter

Image size: 512x512  
FD notch radius: 16

Multiply by  
this, or ...



... convolve  
by this



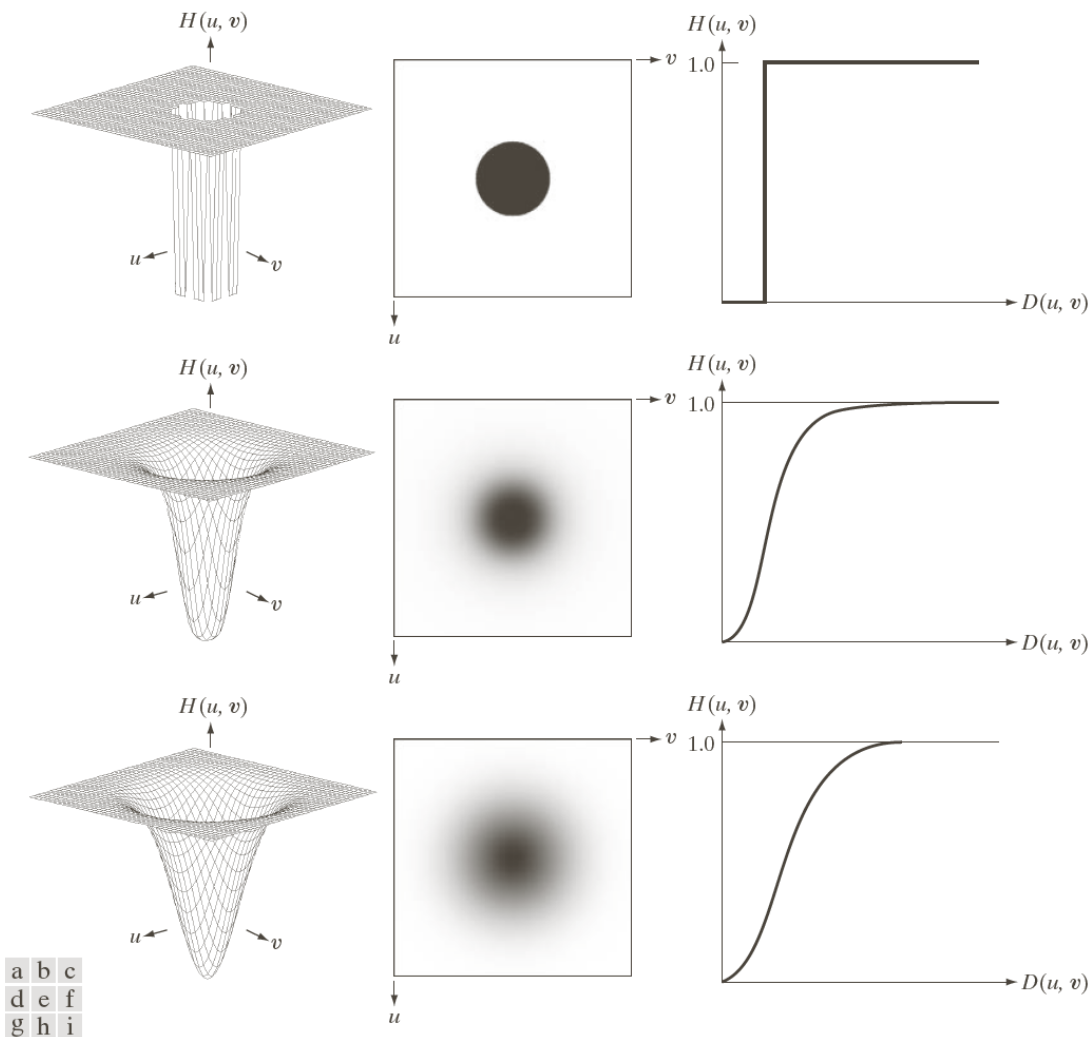
Fourier Domain Rep.

Spatial Representation

Central Profile

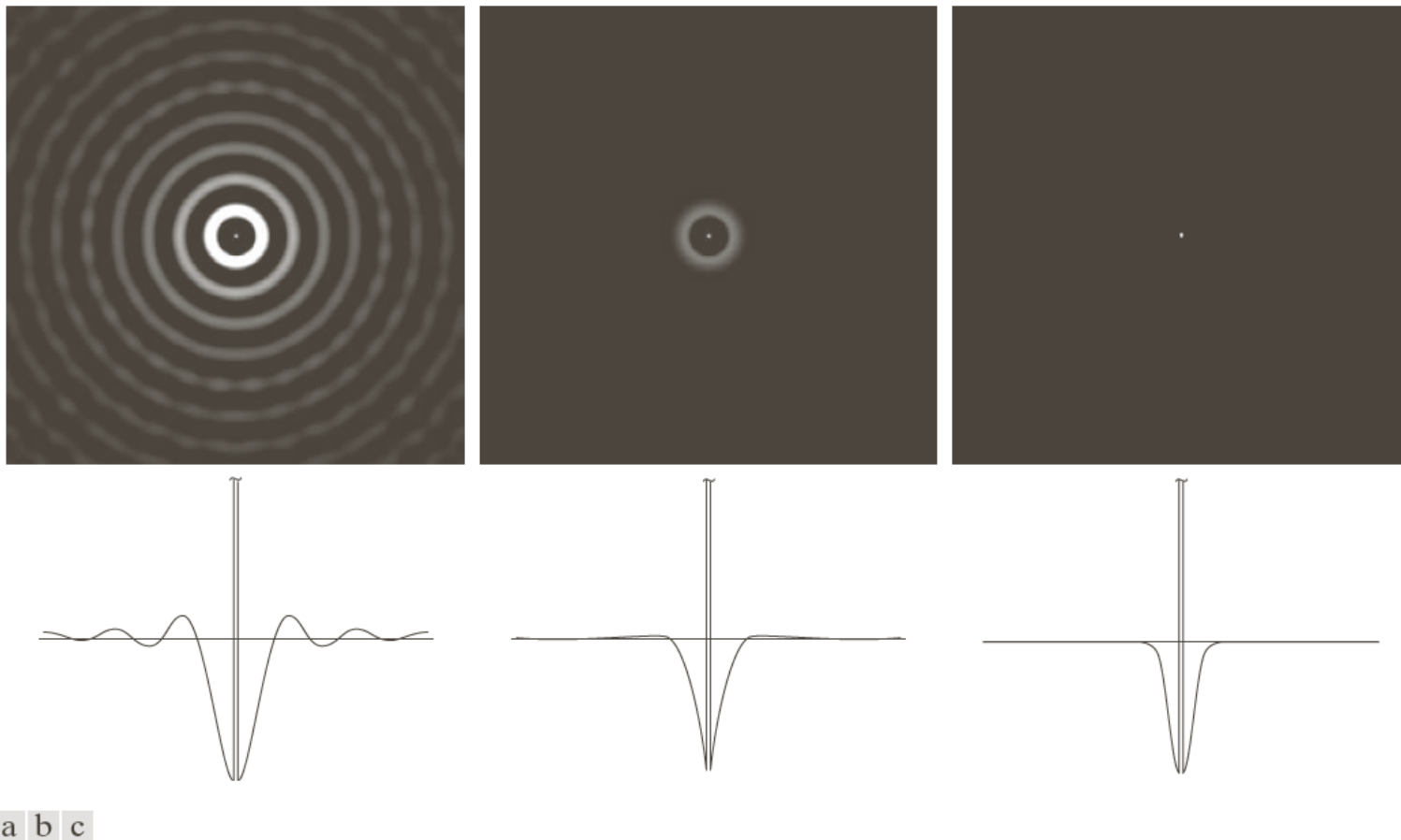
# Highpass filters - 1

$$H_{HPF}(u, v) = 1 - H_{LPF}(u, v)$$



**FIGURE 4.52** Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

# Highpass filters - 2



**FIGURE 4.53** Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

# Highpass filters - 3

Highpass filters.  $D_0$  is the cutoff frequency and  $n$  is the order of the Butterworth filter.

| Ideal                                                                                                 | Butterworth                                  | Gaussian                             |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------|
| $H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$ | $H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$ | $H(u, v) = 1 - e^{-D^2(u,v)/2D_0^2}$ |



# Highpass filters – Results of IHPF



a b c

**FIGURE 4.54** Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with  $D_0 = 30, 60$ , and  $160$ .

# Highpass filters – Results of BHPF



a b c

**FIGURE 4.55** Results of highpass filtering the image in Fig. 4.41(a) using a BHPF of order 2 with  $D_0 = 30, 60,$  and 160, corresponding to the circles in Fig. 4.41(b). These results are much smoother than those obtained with an IHPF.

# Highpass filters – Results of GHPF



a b c

**FIGURE 4.56** Results of highpass filtering the image in Fig. 4.41(a) using a GHPF with  $D_0 = 30, 60$ , and  $160$ , corresponding to the circles in Fig. 4.41(b). Compare with Figs. 4.54 and 4.55.



# Highpass filter application



a b c

**FIGURE 4.57** (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)



# Highpass filter application



**Left:** Original image;

**Right:** Image enhanced using unsharp mask in the spatial domain.



a b

**FIGURE 4.58**

(a) Original, blurry image.

(b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.38(e).

# Band-pass, band-reject filters

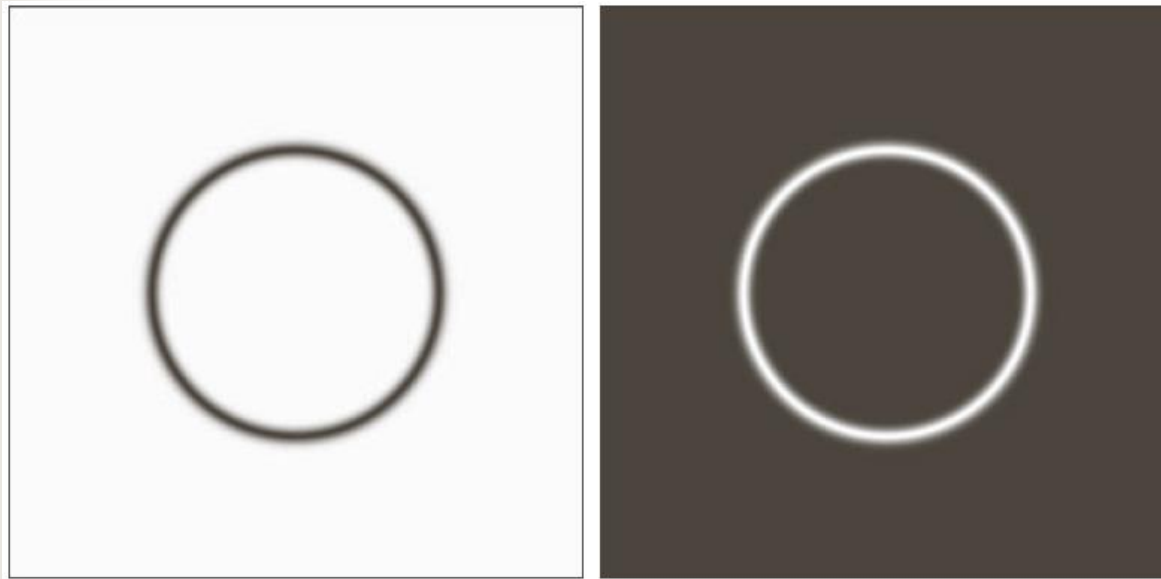
**TABLE 4.6**

Bandreject filters.  $W$  is the width of the band,  $D$  is the distance  $D(u, v)$  from the center of the filter,  $D_0$  is the cutoff frequency, and  $n$  is the order of the Butterworth filter. We show  $D$  instead of  $D(u, v)$  to simplify the notation in the table.

| Ideal                                                                                                                        | Butterworth                                                          | Gaussian                                                     |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------|
| $H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$ | $H(u, v) = \frac{1}{1 + \left[ \frac{DW}{D^2 - D_0^2} \right]^{2n}}$ | $H(u, v) = 1 - e^{-\left[ \frac{D^2 - D_0^2}{DW} \right]^2}$ |

$$H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

# Band-pass, band-reject Gaussian filter

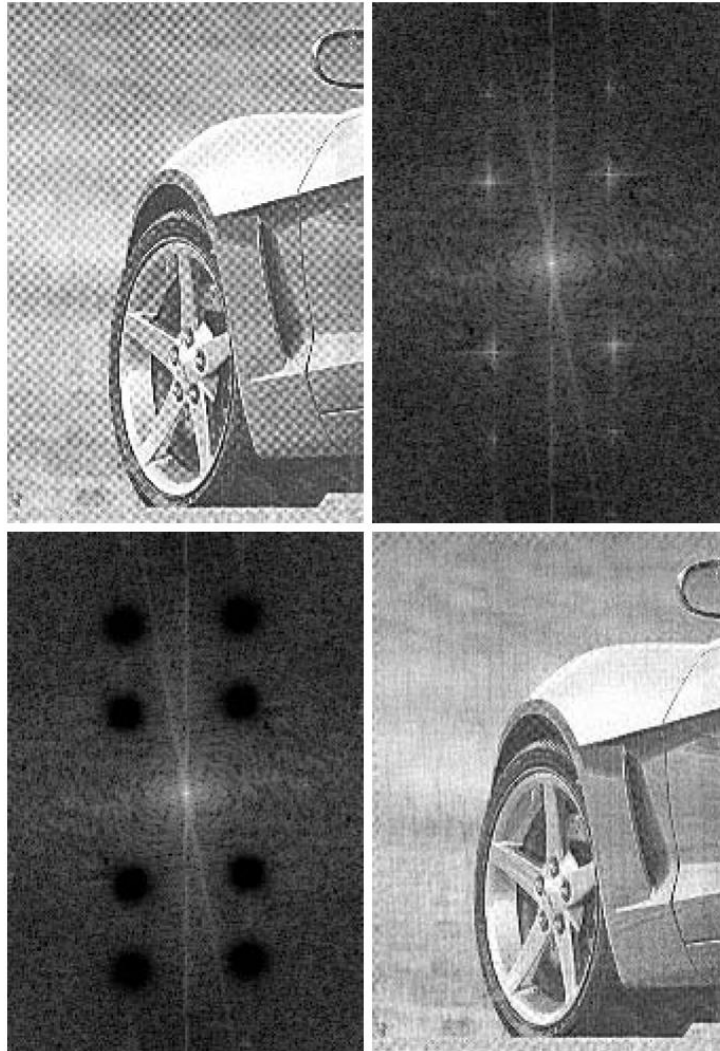


a b

**FIGURE 4.63**

(a) Bandreject Gaussian filter.  
(b) Corresponding bandpass filter.  
The thin black border in (a) was added for clarity; it is not part of the data.

# Example of Band-pass, band-reject filters



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 4.64**

(a) Sampled newspaper image showing a moiré pattern.

(b) Spectrum.

(c) Butterworth notch reject filter multiplied by the Fourier transform.

(d) Filtered image.





# Outline

- Why transform?
- 2D Fourier transform
  - DFT and 2D-DFT
  - properties
  - implementation
  - applications
- Discrete cosine transform (DCT)
  - definition & visualization
  - Implementation
- Multi-resolution and wavelets

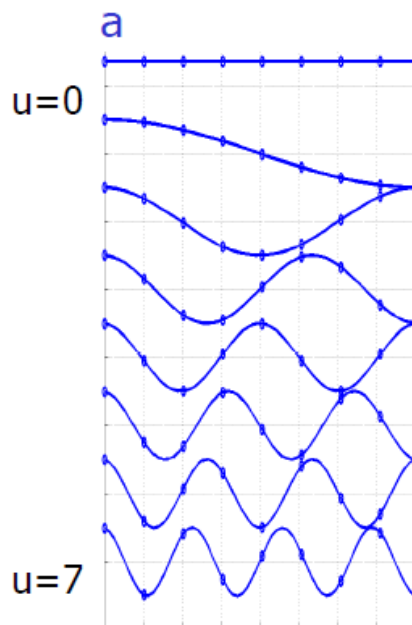
# DCT vs DFT

$$y = Ax$$

1D-DCT

$$a(0, n) = \sqrt{\frac{1}{N}} \quad u = 0$$

$$a(u, n) = \sqrt{\frac{2}{N}} \cos \frac{\pi(2n+1)u}{2N}$$

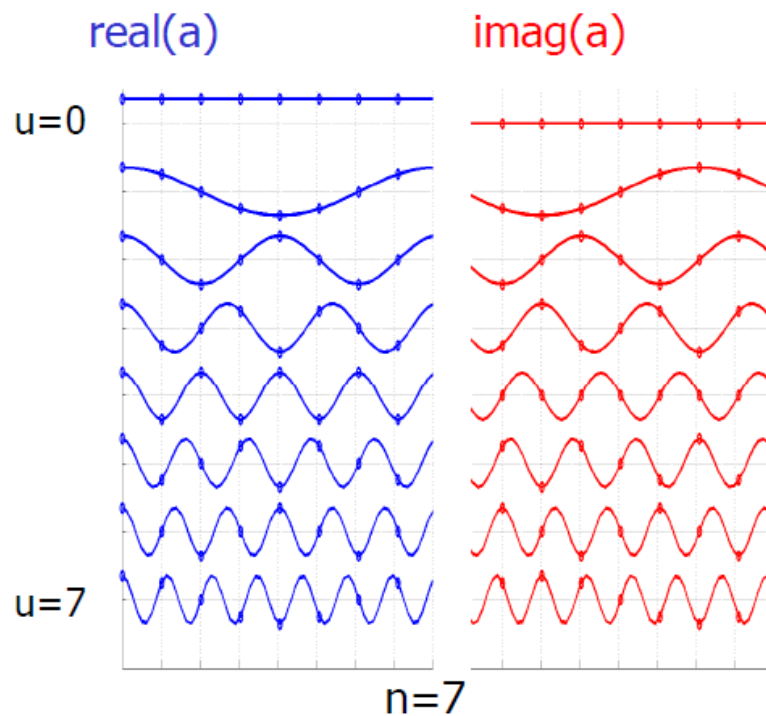


1D-DFT

$$a(u, n) = e^{-j2\pi \frac{un}{N}}$$

$$= \cos(2\pi \frac{un}{N}) + j \sin(2\pi \frac{un}{N})$$

$$u = 1, 2, \dots, N-1$$

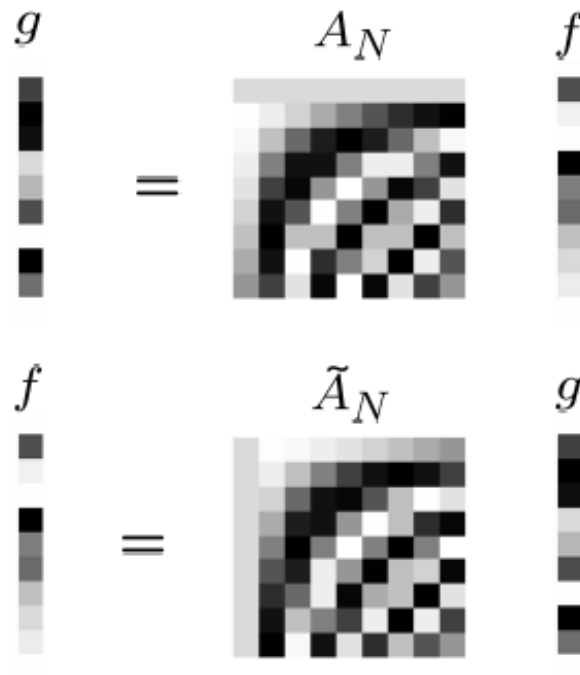


# Transform as basis expansion

$$g(u) = \sum_{n=0}^{N-1} f(n) a_N^{un}$$

$$f(n) = \sum_{u=0}^{N-1} g(u) \tilde{a}_N^{un}$$

inverse  
transform



$$\text{DFT: } a_N^{un} = e^{-j2\pi \frac{un}{N}}, \quad \tilde{a}_N^{un} = a_N^{*un}, \quad \tilde{A}_N = A_N^{*T}$$

$$\text{DCT: } a_N^{0n} = \sqrt{\frac{1}{N}} \quad u = 0$$

$$a_N^{un} = \sqrt{\frac{2}{N}} \cos \frac{\pi(2n+1)u}{2N} \quad u = 1, \dots, N-1$$

$$\tilde{a}_N^{un} = a_N^{un}, \quad \tilde{A}_N = A_N^T$$

# 1-D Discrete Cosine Transform (DCT)

$$\begin{cases} Z(k) = \sum_{n=0}^{N-1} z(n) \cdot \alpha(k) \cos \left[ \frac{\pi(2n+1)k}{2N} \right] \\ z(n) = \sum_{k=0}^{N-1} Z(k) \cdot \alpha(k) \cos \left[ \frac{\pi(2n+1)k}{2N} \right] \end{cases}$$
$$\alpha(0) = \frac{1}{\sqrt{N}}, \alpha(k) = \sqrt{\frac{2}{N}}$$

- Transform matrix  $A$ 
  - $a(k,n) = \alpha(0)$  for  $k=0$
  - $a(k,n) = \alpha(k) \cos[\pi(2n+1)/2N]$  for  $k>0$
- $A$  is real and orthogonal
  - rows of  $A$  form orthonormal basis
  - $A$  is not symmetric!
  - DCT is not the real part of unitary DFT!



# DFT and DCT in Matrix Notations

Matrix notation for 1D transform

$$y = Ax, \quad x = A^{-1}y$$

1D-DCT

$$a(0, n) = \sqrt{\frac{1}{N}} \quad u = 0$$

$$a(u, n) = \sqrt{\frac{2}{N}} \cos \frac{\pi(2n+1)u}{2N} \\ u = 1, 2, \dots, N-1$$

1D-DFT

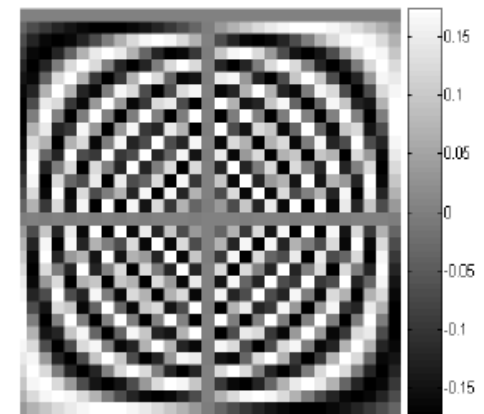
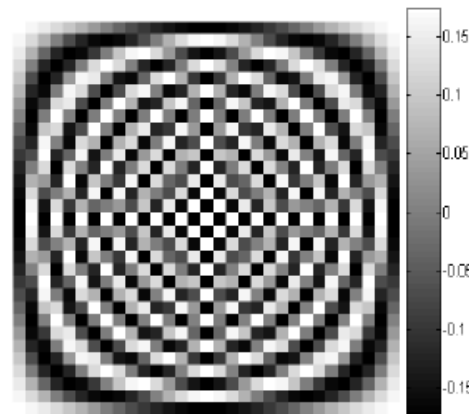
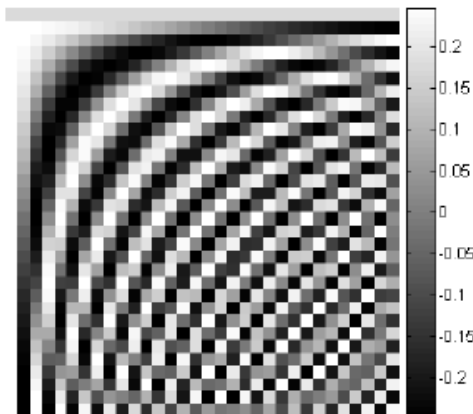
$$a(u, n) = e^{-j2\pi \frac{un}{N}} \\ = \cos(2\pi \frac{un}{N}) - j \sin(2\pi \frac{un}{N})$$

N=32

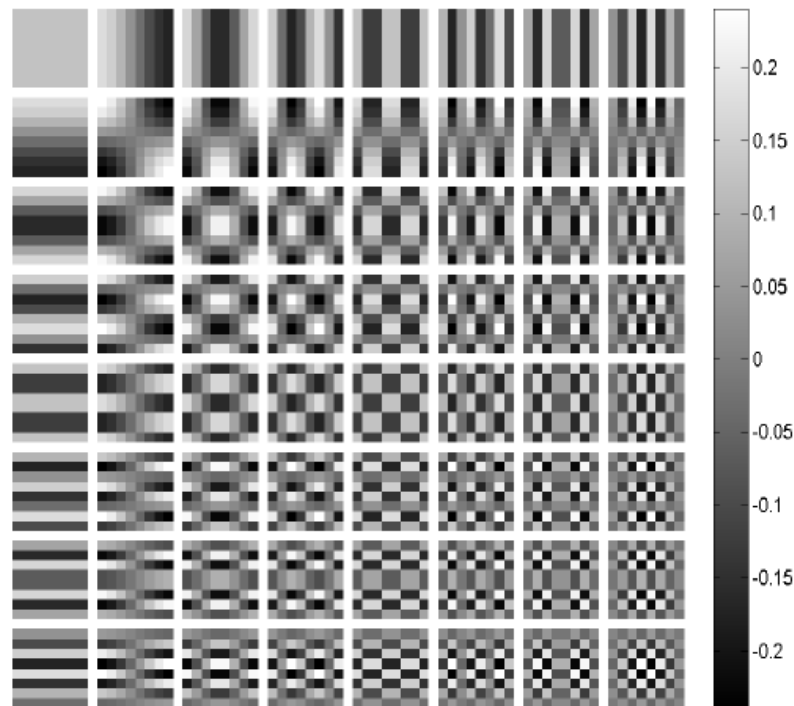
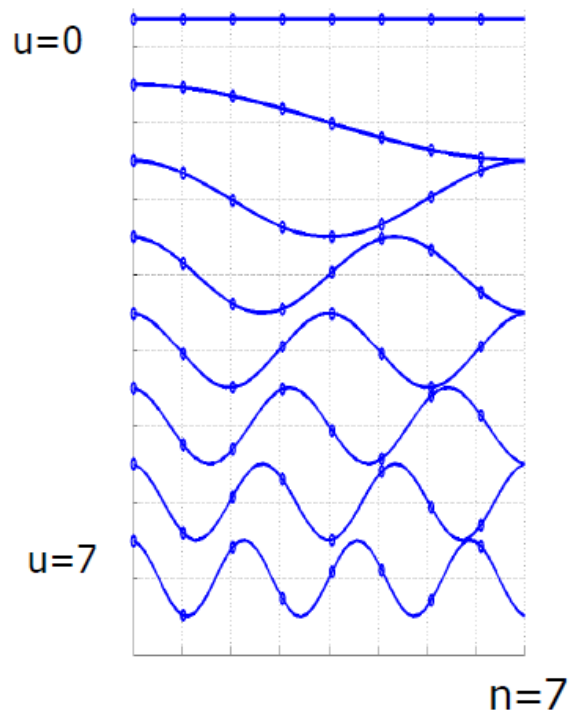
A

real(A)

imag(A)

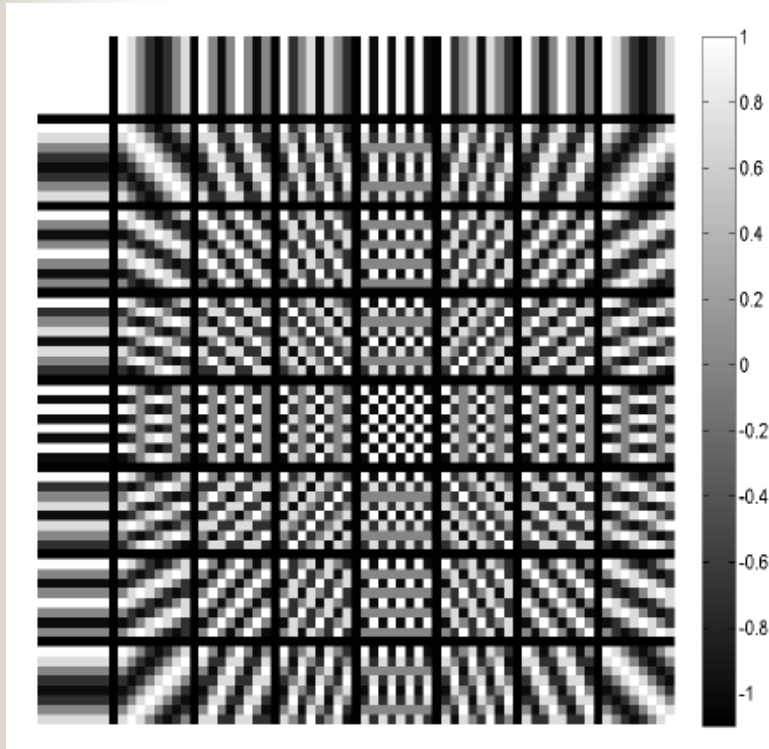


# From 1D-DCT to 2D-DCT

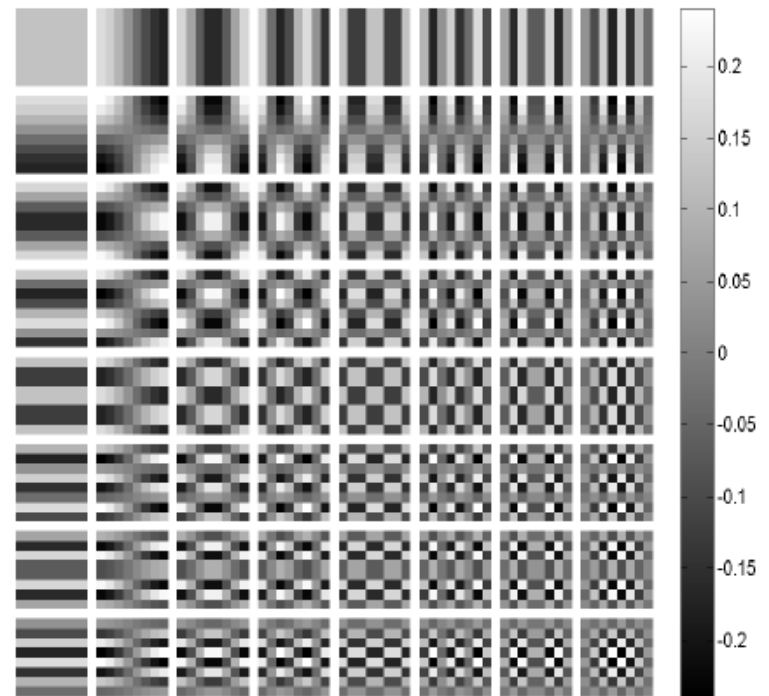


- Rows of  $A$  form a set of orthonormal basis
- $A$  is not symmetric!
- DCT is not the real part of unitary DFT!

# Basis images: DFT (real) vs DCT

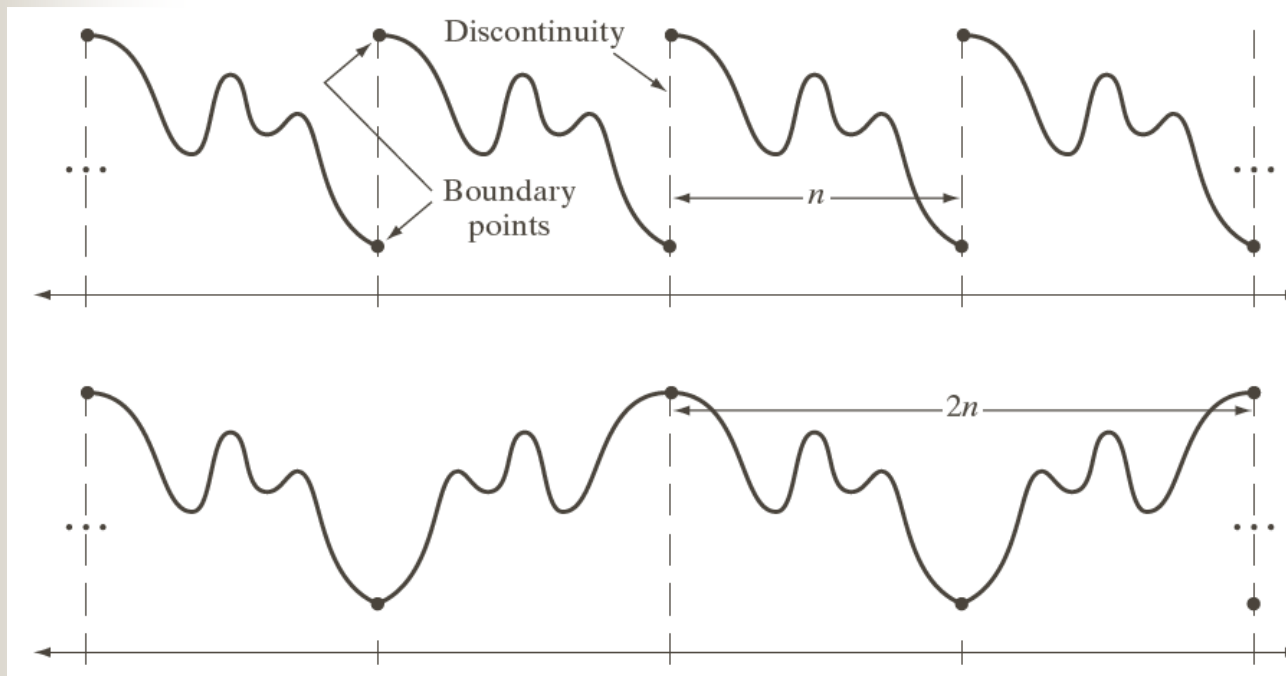


DFT (real)



DCT

# Periodicity Implied by DFT and DCT



**FIGURE 8.25** The periodicity implicit in the 1-D (a) DFT and (b) DCT.

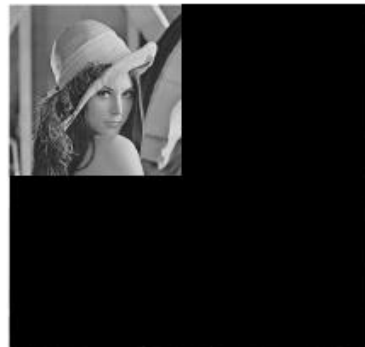
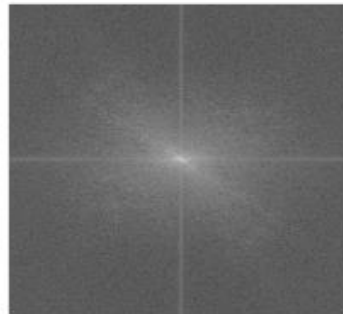
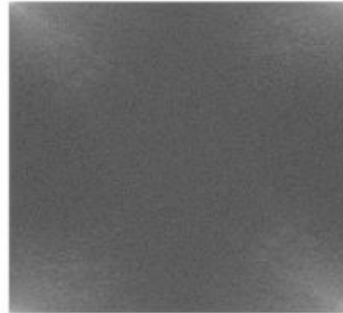


# DFT and DCT on Lena

DFT2

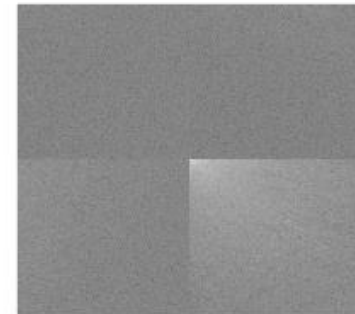
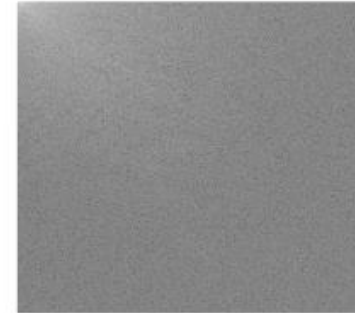


Shift low-freq  
to the center



Assume periodic and zero-padded ...

DCT2



Assume reflection ...

# Using FFT to implement fast DCT

- Reorder odd and even elements

$$\begin{cases} \tilde{z}(n) = z(2n) \\ \tilde{z}(N - n - 1) = z(2n + 1) \end{cases} \quad \text{for } 0 \leq n \leq \frac{N}{2} - 1$$

- Split the DCT sum into odd and even terms

$$\begin{aligned} Z(k) &= \alpha(k) \left\{ \sum_{n=0}^{N/2-1} z(2n) \cdot \cos \left[ \frac{\pi(4n+1)k}{2N} \right] + \sum_{n=0}^{N/2-1} z(2n+1) \cdot \cos \left[ \frac{\pi(4n+3)k}{2N} \right] \right\} \\ &= \alpha(k) \left\{ \sum_{n=0}^{N/2-1} \tilde{z}(n) \cdot \cos \left[ \frac{\pi(4n+1)k}{2N} \right] + \sum_{n=0}^{N/2-1} \tilde{z}(N-n-1) \cdot \cos \left[ \frac{\pi(4n+3)k}{2N} \right] \right\} \\ &= \alpha(k) \left\{ \sum_{n=0}^{N/2-1} \tilde{z}(n) \cdot \cos \left[ \frac{\pi(4n+1)k}{2N} \right] + \sum_{n'=N/2}^{N-1} \tilde{z}(n') \cdot \cos \left[ \frac{\pi(4N-4n'-1)k}{2N} \right] \right\} \\ &= \alpha(k) \sum_{n=0}^{N-1} \tilde{z}(n) \cdot \cos \left[ \frac{\pi(4n+1)k}{2N} \right] = \text{Re} \left[ \alpha(k) e^{-j\pi k/2N} \sum_{n=0}^{N-1} \tilde{z}(n) \cdot e^{-j2\pi nk/N} \right] \\ &= \text{Re} \left[ \alpha(k) e^{-j\pi k/2N} \text{DFT} \{ \tilde{z}(n) \}_N \right] \end{aligned}$$



# The Desirables for Image Transforms

|                                                         | DFT | DCT |
|---------------------------------------------------------|-----|-----|
| ■ Theory                                                |     |     |
| ■ Inverse transform available                           | ✓   | ✓   |
| ■ Energy conservation (Parsevell)                       | ✓   | ✓   |
| ■ Good for compacting energy                            | ?   | ?   |
| ■ Orthonormal, complete basis                           | ✓   | ✓   |
| ■ (sort of) shift- and rotation invariant               | ✓   | ✓   |
| ■ Implementation                                        |     |     |
| ■ Real-valued                                           | x   | ✓   |
| ■ Separable                                             | ✓   | ✓   |
| ■ Fast to compute w. butterfly-like structure           | ✓   | ✓   |
| ■ Same implementation for forward and inverse transform | ✓   | ✓   |
| ■ Application                                           |     |     |
| ■ Useful for image enhancement                          | ✓   |     |
| ■ Capture perceptually meaningful structures in images  | ✓   |     |



# Outline

- Why transform?
- 2D Fourier transform
  - a picture book for DFT and 2D-DFT
  - properties
  - implementation
  - applications
- Discrete cosine transform (DCT)
  - definition & visualization
  - Implementation
- Multi-resolution and wavelets





# Fourier Transform vs Wavelet Transform

## ■ Similarities

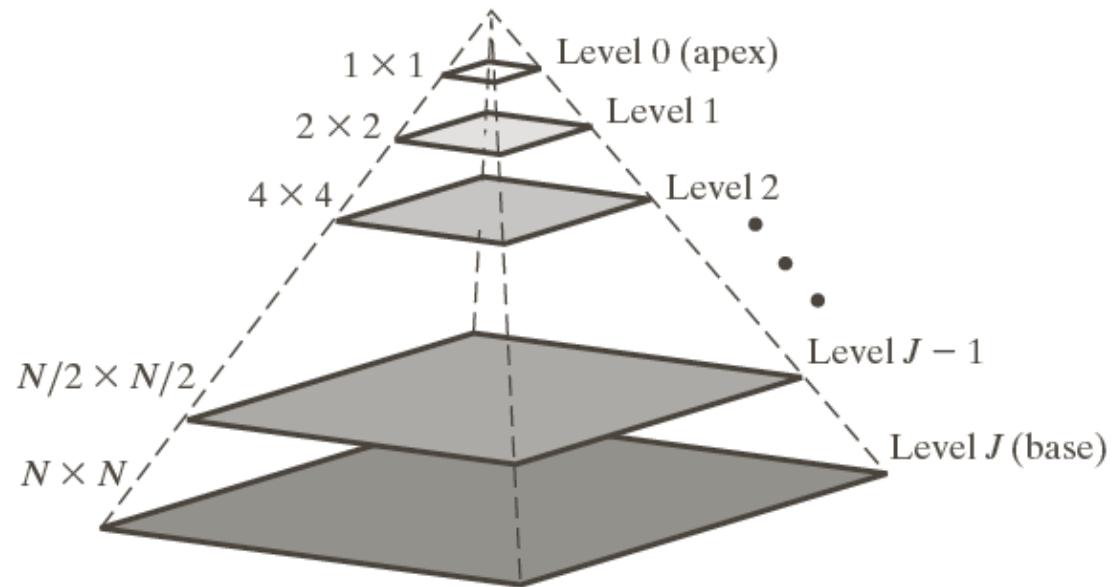
- both are linear operations.
- The inverse transform matrix for both is the transpose of the original.
- The basis functions are localized in frequency

## ■ Dissimilarities

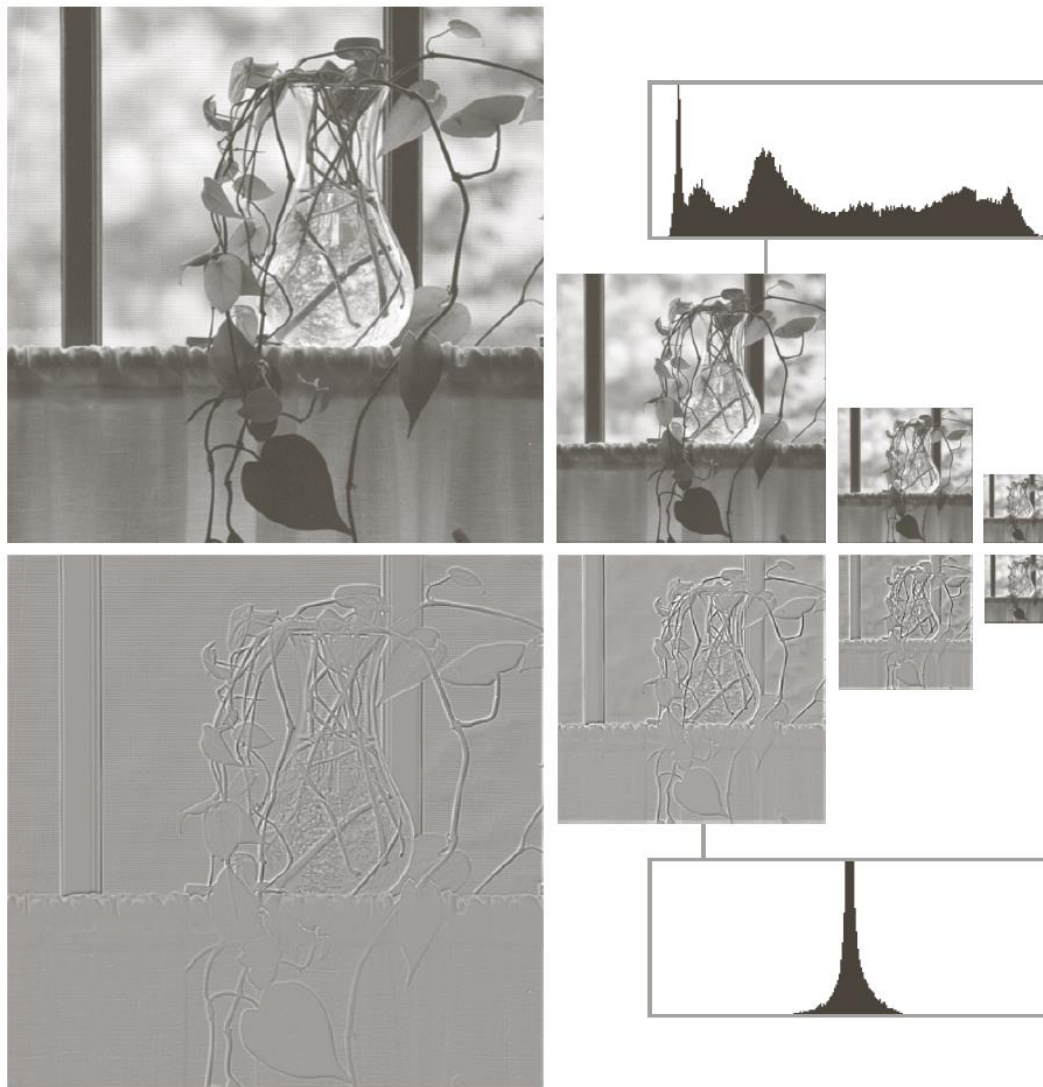
- The basis functions for DFT are sinusoids
- The basis functions for DWT are small waves.
- DFT only provides frequency information
- DWT provides both frequency and temporal information
- DWT is multiresolution process.

# Multi-resolution (MR) and Image Pyramids

- Multi-resolution – represent and analyze signals (images) at more than one resolution.
- Image Pyramids -- represent images at more than one resolution.



# Example: Image Pyramids



a

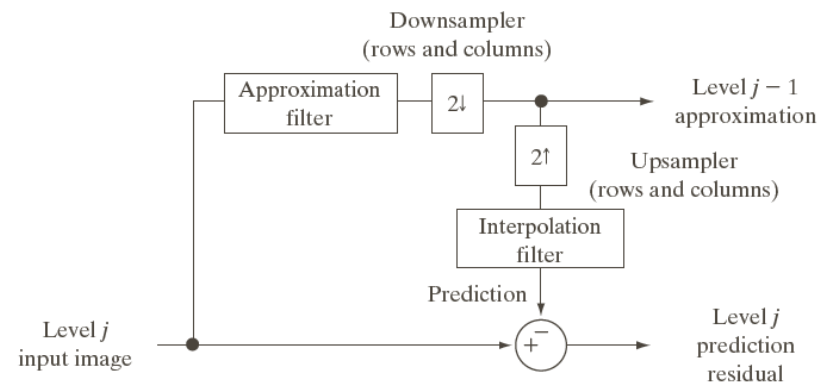
b

**FIGURE 7.3**

Two image pyramids and their histograms:  
(a) an approximation pyramid;  
(b) a prediction residual pyramid.

# Steps to Construct the Image Pyramid

1. Given an image at level  $j$
2. Filter the input and downsample the filtered result by a factor of 2; This gives the image at level  $j-1$
3. Goto 1
4. Upsample and filter the image at level  $j-1$ ; this gives an approximation of the image at level  $j$
5. Subtract this result from the image at level  $j$ ; this give the prediction residual at level  $j$
6. Goto 1



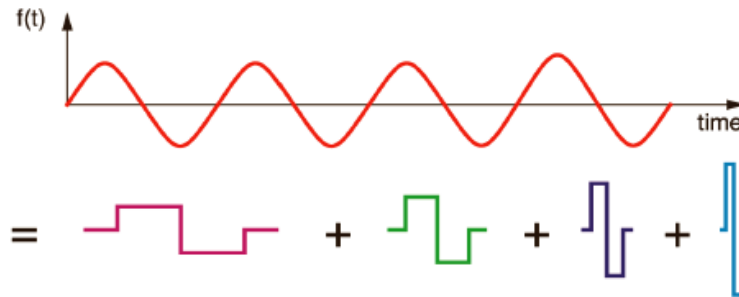




# Typical Filters

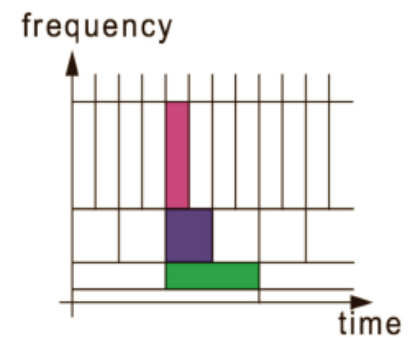
- For the multiresolution pyramid, we use spatial filters:
  - Neighborhood averaging
  - Lowpass Gaussian filter
- For the residual pyramid, we use interpolation filters:
  - bilinear
  - bicubic

**1910: Alfred Haar discovers the Haar wavelet**  
 “dual” to the Fourier construction



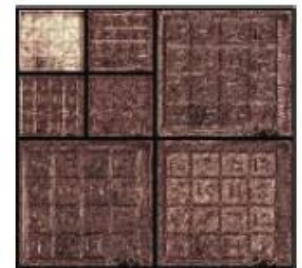
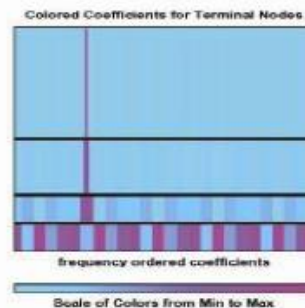
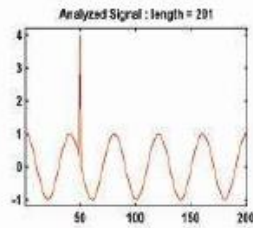
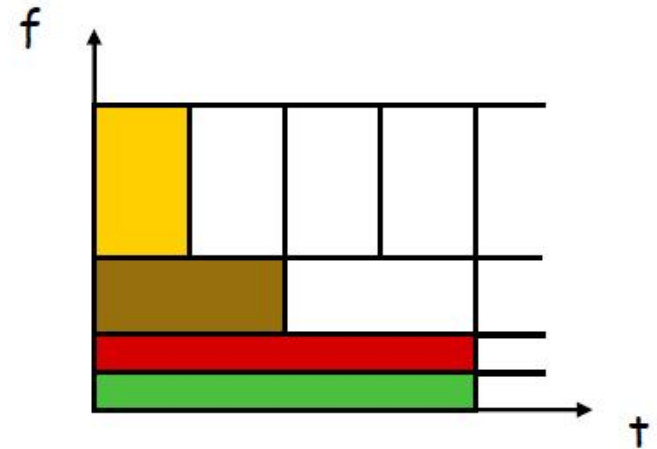
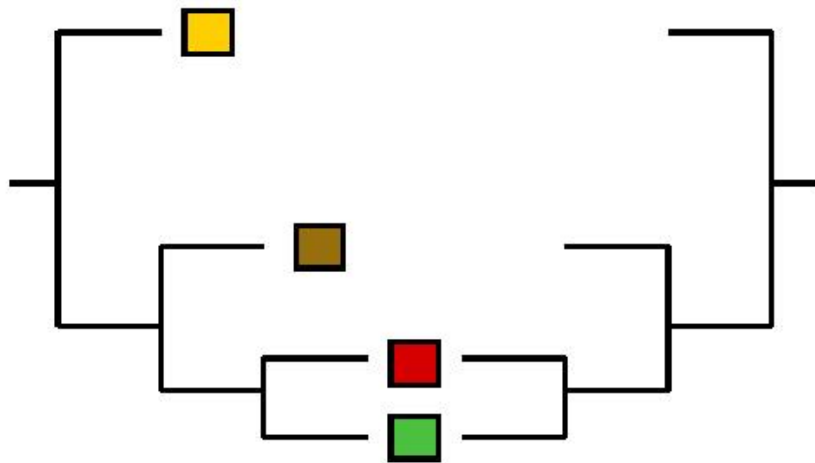
**Haar series:**

- Scale changes  $S_0, 2S_0, 4S_0, 8S_0 \dots$
- orthogonality



# Discrete Wavelet Transform

- Iterate only on the lowpass channel



# 2D Discrete Wavelet Transform

$$\begin{aligned}\varphi(x, y) &= \varphi(x)\varphi(y) & \psi^H(x, y) &= \psi(x)\varphi(y) \\ \psi^V(x, y) &= \varphi(x)\psi(y) \\ \psi^D(x, y) &= \psi(x)\psi(y)\end{aligned}$$

$$\varphi_{j,m,n}(x, y) = 2^{\frac{j}{2}} \varphi(2^j x - m, 2^j y - n)$$

$$\psi_{j,m,n}^i(x, y) = 2^{\frac{j}{2}} \psi^i(2^j x - m, 2^j y - n), \quad i = H, V, D$$

$$W_{\varphi}(j, m, n) = \frac{1}{\sqrt{MN}} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \varphi_{j,m,n}(x, y)$$

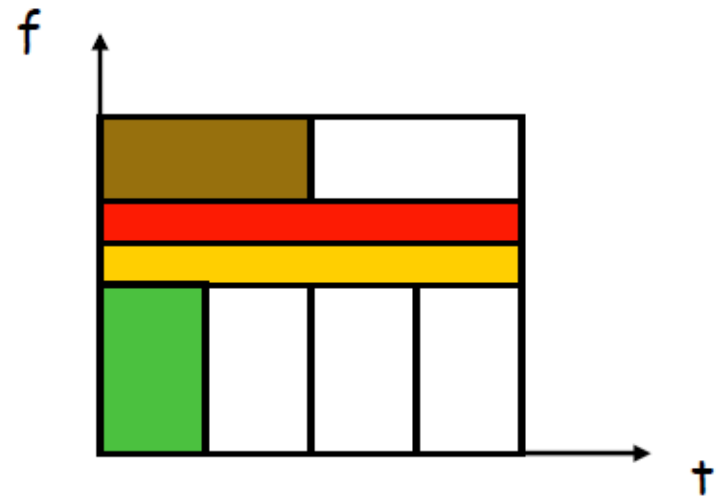
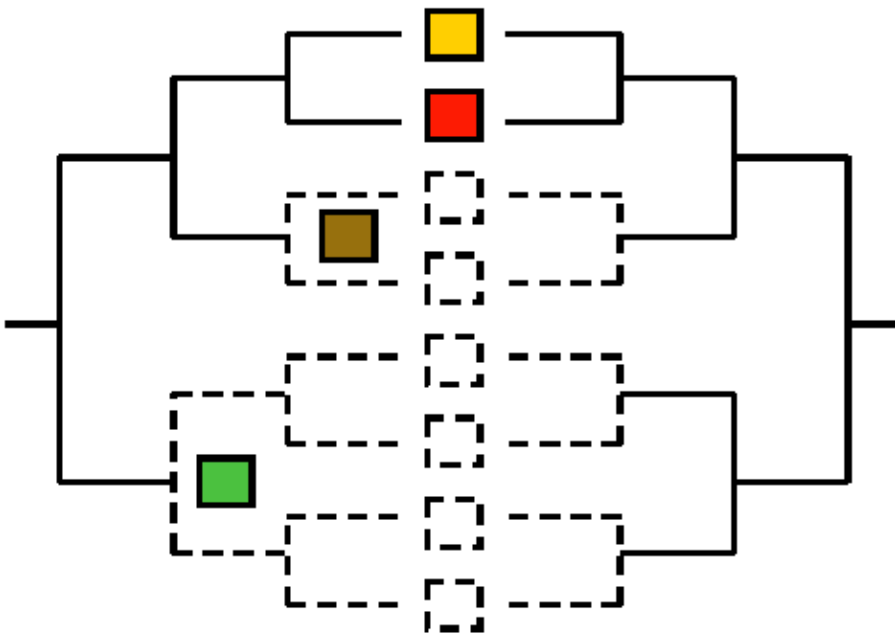
$$W_{\psi}(j, m, n) = \frac{1}{\sqrt{MN}} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \psi_{j,m,n}(x, y)$$

$$\begin{aligned}f(x, y) &= \frac{1}{\sqrt{MN}} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{\varphi}(j_0, m, n) \varphi_{j_0,m,n}(x, y) \\ &\quad + \frac{1}{\sqrt{MN}} \sum_{i=H,V,D} \sum_{j=j_0}^{\infty} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{\psi}^i(j, m, n) \psi_{j,m,n}^i(x, y)\end{aligned}$$



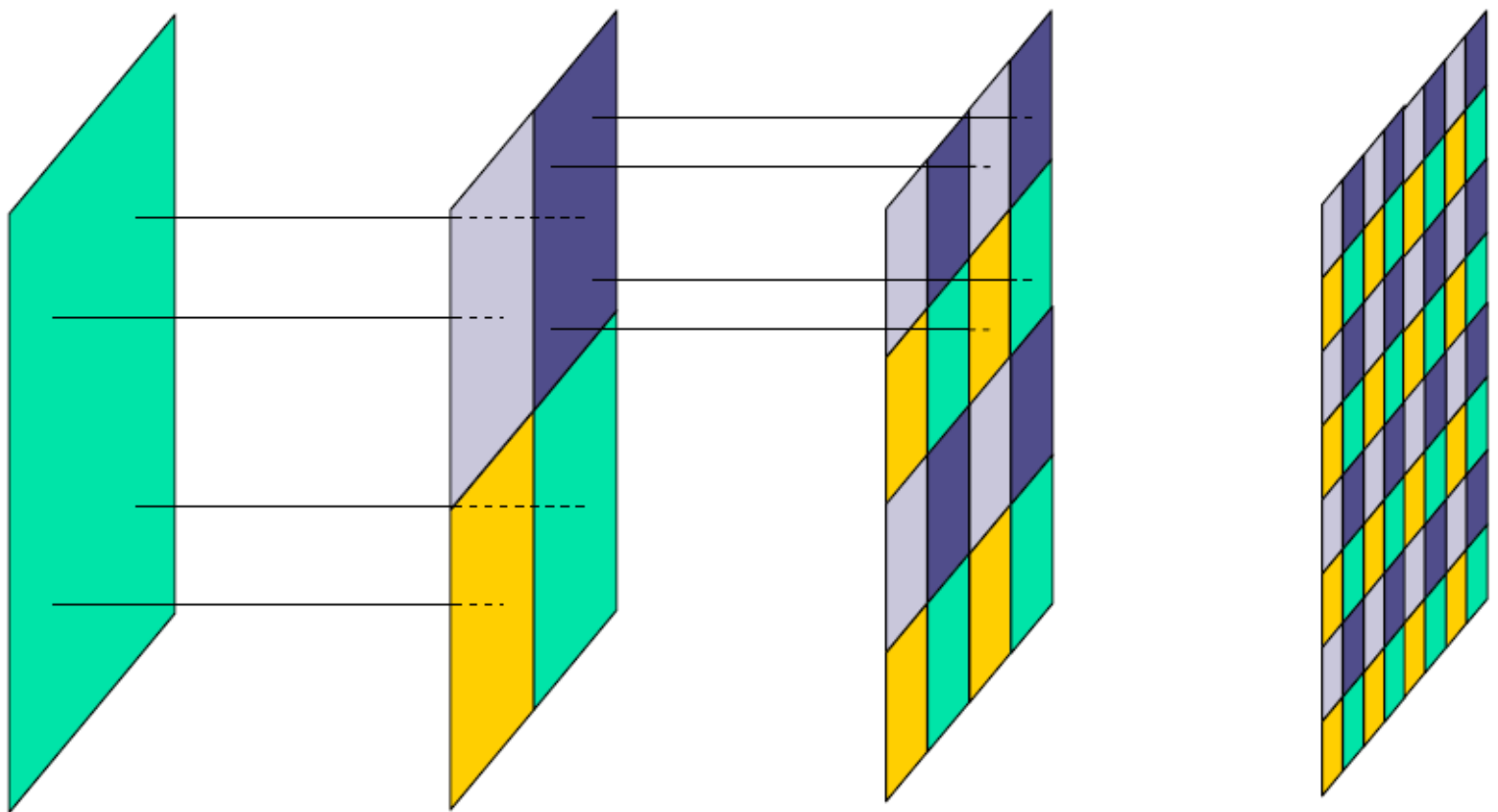
# Wavelet packet

- Iterate on both the low pass and (selected) high-pass channels



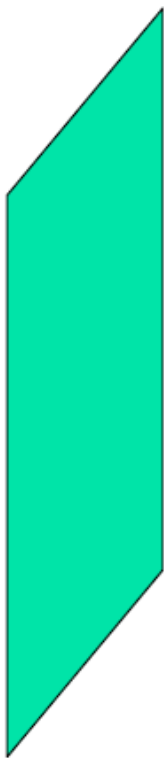
# Wavelet packet

- First stage: full decomposition



# Wavelet packet

- Second stage: pruning



$\text{Cost}(\text{parent}) < \text{Cost}(\text{children})$



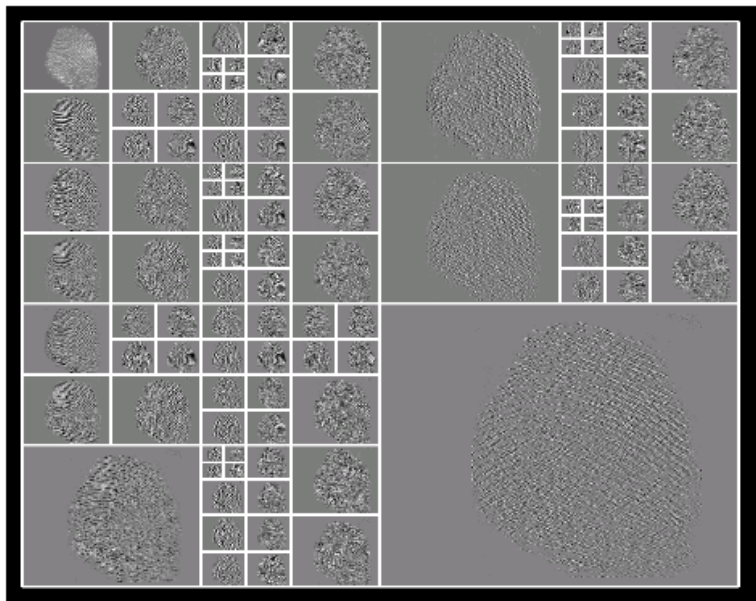
# Feature extraction

|               |          |               |               |
|---------------|----------|---------------|---------------|
| L1<br>LL      | L1<br>HL | Level 2<br>HL | Level 3<br>HL |
| L1<br>LH      | L1<br>HH |               |               |
| Level 2<br>LH |          | Level 2<br>HH |               |
| Level 3<br>LH |          |               | Level 3<br>HH |

Wavelet representation of an image

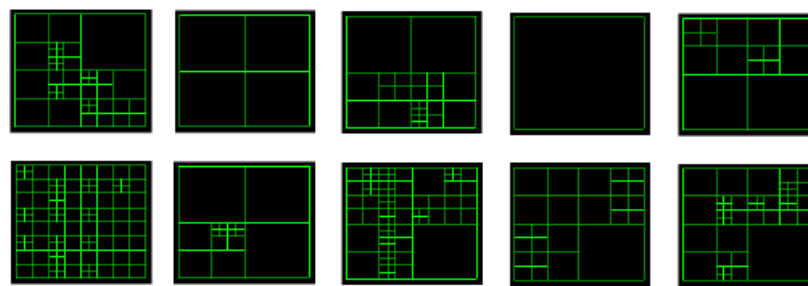


# Fingerprint feature extraction

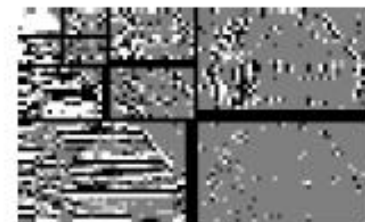
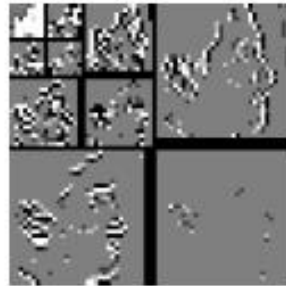
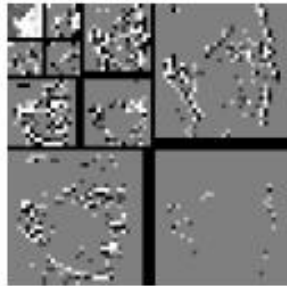


- MR system
  - Introduces adaptivity
  - Template matching performed on different space-frequency regions
  - Builds a different decomposition for each class

$$F(\text{parent}) > F(\text{child 1}) + F(\text{child 2}) + F(\text{child 3}) + F(\text{child 4}), \quad F = \frac{1}{E}$$



# Face & car feature extraction



Images and their wavelet transforms

# Object Detection – faces & cars



Figure 1. Examples of computer detection of human faces



Figure 2. Examples of computer detection of automobiles

Results produced by Henry Schneiderman.





# Announcement

- HW2 is due: 3/3.
- Ali will present HW2 on 3/4.
- HW3 is out 3/4, due: 3/24.
- Final Project starts on 3/25.
- **Midterm Exam: Mar. 11, 2026.**